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Fig 5.1 Our understanding of hormones, while still evolving, originated from a mix of random experiments, coincidental findings, and extraordinary sounding characters! One of these was the 'castrati' that featured in opera throughout the 16th, 17th, and 18th centuries. These were boys who were castrated before puberty. The voice of a castrato was pure and forceful, due to their enormous lung capacity and resulting breath control. They also experienced no temporal recession, and their arms and legs were long. One of the most well renowned was Farinelli, the stage name of Carlo Maria Michelangelo Nicola Broschi. It was said he had a well-modulated soprano voice with extraordinary breath control. His picture hangs in Handel's house in London. The practice was stopped in the early 20th century when it was acknowledged how inhumane the operation was.

The essence of endocrinology

For scientists

- Define a syndrome, and match it to a gland malfunction.
- Measure the gland's output in the peripheral blood. Define clinical syndromes associated with too much or too little secretion (*hyper-* and *hypo-*syndromes, respectively; *eu-* means normal, neither ↑ nor ↓, as in *euthyroid*). Note factors that may make measurement variable, eg diurnal release of cortisol.
- If suspecting hormone deficiency, test by stimulating the gland that produces it (eg short ACTH stimulation test or *Synacthen*® test in Addison's). If the gland is not functioning normally, there will be a blunted response to stimulation.
- If suspecting hormone excess, test by inhibiting the gland that produces it (eg dexamethasone suppression test in Cushing's). If there is a hormone-secreting tumour then this will fail to suppress via normal feedback mechanisms.
- Find a way to image the gland. NB: non-functioning tumours or 'incidentalomas' may be found in health, see p224. Imaging alone does not make the diagnosis.
- Aim to halt disease progression; diet and exercise can stop progression of impaired fasting glucose to frank diabetes.^{1,2} For other glands, halting progression will depend on understanding autoimmunity, and the interaction of genes and environment. In thyroid autoimmunity (an archetypal autoimmune disease), it is possible to track interactions between genes and environment (eg smoking and stress) via expression of immunologically active molecules (HLA class I and II, adhesion molecules, cytokines, CD40, and complement regulatory proteins).³

Endocrinologists love this reductionist approach, but have been less successful at understanding *emergent phenomena*—those properties and performances of ours that cannot be predicted from full knowledge of our perturbed parts. We understand the diurnal nature of cortisol secretion, for example, but the science of relating this to dreams, the consolidation of memory, and the psychopathology of families and other groups (such as the endocrinology ward round you may be about to join) is in its infancy. But as doctors we are steeped in the hormonal lives of patients (as they are in ours)—and we may as well start by recognizing this now.

For those doing exams

'What's wrong with *him*?' your examiner asks, boldly. While you apologize to the patient for this rudeness by asking, 'Is it alright if we speak about you as if you weren't here?', think to yourself that if you were a betting man or woman you would wager that the diagnosis will be endocrinological. In no other discipline are gestalt impressions so characteristic. To get good at recognizing these conditions, spend time in endocrinology outpatients and looking at collections of clinical photographs. Also, specific cutaneous signs are important, as follows.

Thyrotoxicosis: Hair loss; pretibial myxoedema (confusing term, p218); onycholysis (nail separation from the nailbed); bulging eyes (exophthalmos/proptosis).

Hypothyroidism: Hair loss; eyebrow loss; cold, pale skin; characteristic face. You might, perhaps should, fail your exam if you blurt out 'Toad-like face'.

Cushing's syndrome: Central obesity and wasted limbs (= 'lemon on sticks' see fig 5.2); moon face; buffalo hump; supraclavicular fat pads; striae.

Addison's disease: Hyperpigmentation (face, neck, palmar creases).

Acromegaly: Acral (distal) + soft tissue overgrowth; big jaws (macrognathia), hands and feet; the skin is thick; facial features are coarse.

Hyperandrogenism (♀): Hirsutism; temporal balding; acne.

Hypopituitarism: Pale or yellow tinged thinned skin, resulting in fine wrinkling around the eyes and mouth, making the patient look older.

Hypoparathyroidism: Dry, scaly, puffy skin; brittle nails; coarse hair.

Pseudohypoparathyroidism: Short stature, short neck, and short 4th and 5th metacarpals.



Fig 5.2 'Lemon on sticks.'

Hormones are chemical messengers which act directly on nearby cells (paracrine effect), on the cell of origin (autocrine effect), at a distant site (endocrine effect), or as neurotransmitters (brain and gastrointestinal tract). Thirst, thermal regulation, appetite, sleep cycles, menstrual cycle, and stress/mood are all controlled by the hypothalamus. Releasing factors produced by the hypothalamus reach the pituitary via the portal system (pituitary stalk), see **fig 5.3**. The releasing factors stimulate or inhibit the production of hormones from the anterior pituitary, **fig 5.4**. Vasopressin and oxytocin are produced in the hypothalamus and stored and released from the posterior pituitary.

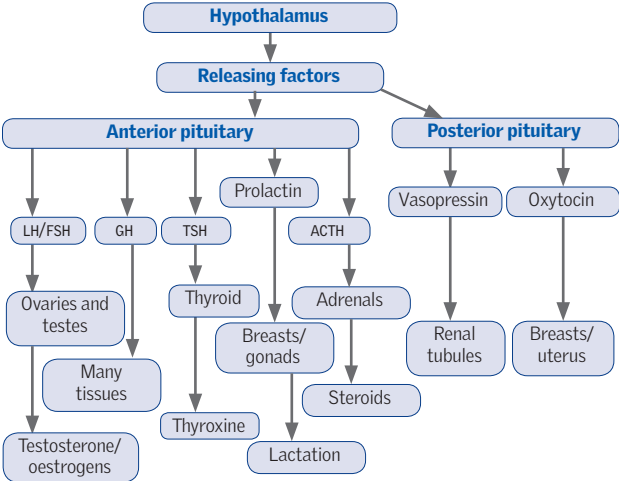


Fig 5.3 Hypothalamic-pituitary axis.

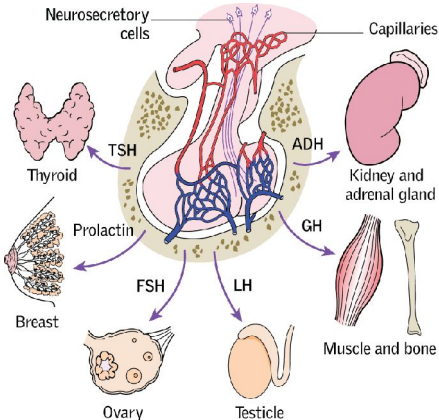


Fig 5.4 Neuroregulation and integration of endocrine axes makes us who we are—and who we are and what we do feeds back into our hormonal milieu. Multifactorial disruptions within the growth hormone (GH), luteinizing hormone (LH)–testosterone, adrenocorticotropin (ACTH)–cortisol and insulin axes play a major role in healthy maturation and ageing.



DM results from lack, or reduced effectiveness, of endogenous insulin. Hyperglycaemia is one aspect of a far-reaching metabolic derangement, which causes serious microvascular (retinopathy, nephropathy, neuropathy) or macrovascular problems: stroke, MI, renovascular disease, limb ischaemia. *So think of DM as a vascular disease:*¹ adopt a holistic approach and consider other cardiovascular risk factors too.

Categories of diabetes and dyslipidaemia

Type 1 DM: Usually adolescent onset but may occur at *any* age. *Cause:* insulin deficiency from autoimmune destruction of insulin-secreting pancreatic β cells. Patients must have insulin, and are prone to ketoacidosis and weight loss. Associated with other autoimmune diseases ($>90\%$ HLA DR3 $\pm$ DR4). Concordance is only $\sim 30\%$ in identical twins, indicating environmental influence. Four genes are important: one (6q) determines islet sensitivity to damage (eg from viruses or cross-reactivity from cows' milk-induced antibodies). Latent autoimmune diabetes of adults (LADA) is a form of type 1 DM, with slower progression to insulin dependence in later life.

Type 2 DM: (Formerly non-insulin-dependent DM, NIDDM) is at 'epidemic' levels in many places, mainly due to changes in lifestyle, but also because of better diagnosis and improved longevity.⁴ Higher prevalence in Asians, men, and the elderly (up to 18%). Most are over 40yrs, but teenagers are now getting type 2 DM (OHCS p156). *Cause:* $\downarrow$ insulin secretion $\pm$ $\uparrow$ insulin resistance. It is associated with obesity, lack of exercise, calorie and alcohol excess. $\geq 80\%$ concordance in identical twins, indicating stronger genetic influence than in type 1 DM. Typically progresses from a preliminary phase of impaired glucose tolerance (IGT) or impaired fasting glucose (IFG). (► This is a unique window for lifestyle intervention.) Maturity onset diabetes of the young (MODY) is a rare autosomal dominant form of type 2 DM affecting young people.

Impaired glucose tolerance (IGT): Fasting plasma glucose $<7\text{mmol/L}$ and OGTT (oral glucose tolerance) 2h glucose $\geq 7.8\text{mmol/L}$ but $<11.1\text{mmol/L}$.

Impaired fasting glucose (IFG): Fasting plasma glucose $\geq 6.1\text{mmol/L}$ but $<7\text{mmol/L}$ (WHO criteria). Do an OGTT to exclude DM. The cut-off point is somewhat arbitrary. IGT and IFG denote different abnormalities of glucose regulation (post-prandial and fasting). There may be lower risk of progression to DM in IFG than IGT. Manage both with lifestyle advice (p93) + annual review. Incidence of DM if IFG and HbA1c at high end of normal (37–46mmol/mol) is $\sim 25\%$.⁵

Other causes of DM ► Steroids; anti-HIV drugs; newer antipsychotics.

- Pancreatic: pancreatitis; surgery (where $>90\%$ pancreas is removed); trauma; pancreatic destruction (haemochromatosis, cystic fibrosis); pancreatic cancer.
- Cushing's disease; acromegaly; pheochromocytoma; hyperthyroidism; pregnancy.
- Others: congenital lipodystrophy; glycogen storage diseases.

Metabolic syndrome (syndrome X) Definition from International Diabetes Federation: central obesity (BMI >30 , or $\uparrow$ waist circ, ethnic-specific values), plus two of BP $\geq 130/85$, triglycerides $\geq 1.7\text{mmol/L}$, HDL $\leq 1.03\sigma/1.29\phi\text{mmol/L}$, fasting glucose $\geq 5.6\text{mmol/L}$ or type 2 DM. $\sim 20\%$ are affected; weight, genetics, and insulin resistance important in aetiology. Vascular events—but probably not beyond the combined effect of individual risk factors. R_x : Exercise; $\downarrow$ weight; treat individual components.

¹ Chicken or egg? Most type 2 diabetes-associated genes have a function in the vasculature, and stress in β -cells can result from vascular defects in the pancreas, so maybe vascular events trigger DM.⁴

Diagnosis of diabetes mellitus: WHO criteria (rather arbitrary!)

- Symptoms of hyperglycaemia (eg polyuria, polydipsia, unexplained weight loss, visual blurring, genital thrush, lethargy) AND raised venous glucose detected once—fasting ≥ 7 mmol/L or random ≥ 11.1 mmol/L OR
 - Raised venous glucose on two separate occasions—fasting ≥ 7 mmol/L, random ≥ 11.1 mmol/L or oral glucose tolerance test (OGTT)—2h value ≥ 11.1 mmol/L
 - HbA_{1c} ≥ 48 mmol/mol. Avoid in pregnancy, children, type 1 DM, and haemoglobinopathies.
- Whenever you have a needle in a vein, do a blood glucose (unless recently done); note if fasting or not. Non-systematic, but better than urine tests (false –ves).

Differentiating type 1 and 2 diabetes

Occasionally it may be difficult to differentiate whether a patient has type 1 or 2 DM, although they can present differently (see [table 5.1](#)). Features of type 1 include weight loss; persistent hyperglycaemia despite diet and medications; presence of autoantibodies: islet cell antibodies (ICA) and anti-glutamic acid decarboxylase (GAD) antibodies; ketonuria.

Table 5.1 Differences between type 1 and type 2 diabetes

	Type 1 DM	Type 2 DM
Epidemiology	Often starts before puberty	Older patients (usually)
Genetics	HLA D3 and D4 linked	No HLA association
Cause	Autoimmune β -cell destruction	Insulin resistance/ β -cell dysfunction
Presentation	Polydipsia, polyuria, $\downarrow$ weight, ketosis	Asymptomatic/complications, eg MI

► Not all new-onset DM in older people is type 2: if ketotic $\pm$ a poor response to oral hypoglycaemics (and patient is slim or has a family or personal history of autoimmunity), think of latent autoimmune diabetes in adults (LADA) and measure islet cell antibodies.

What is the best diet for obese patients with type 2 diabetes?

Dietary carbohydrate is a big determinant of postprandial glucose levels, and low-carbohydrate diets improve glycaemic control. How do low-carbohydrate, ketogenic diets (<20g of carbohydrate daily; LCKD) compare with low-glycaemic index, reduced-calorie diet (eg 500kcal/day deficit from weight maintenance diet)? In one randomized study over 24 weeks, LCKD had greater improvements in HbA_{1c} (-15 vs -5 mmol/L), weight (-11 kg vs -7 kg), and HDL. Diabetes drugs were reduced or eliminated in 95% of LCKD vs 62% of LGID participants.⁶ NB: effects on renal function and mortality are unknown so these diets remain controversial.

Monitoring glucose control

- 1 Fingerprick glucose if on insulin (type 1/2). NB: *before* a meal informs about long-acting insulin doses; *after* meals inform about the dose of short-acting insulin.
- 2 Glycated haemoglobin (HbA_{1c}) relates to mean glucose level over previous 8wks (RBC $t_{1/2}$). Targets are negotiable, eg 48–57mmol/mol (depends on patient's wish and arterial risk, eg past MI or stroke). If at risk from the effects of hypoglycaemia, eg elderly patients prone to falls, consider less tight control. Tight control may not alter all-cause mortality.⁷ Complications rise with rising HbA_{1c}, so any improvement helps.
- 3 Be sure to ask about hypoglycaemic attacks (and whether symptomatic). Hypoglycaemic awareness may diminish if control is too tight, or with time in type 1 DM, due to $\downarrow$ glucagon secretion. It may return if control is loosened.

General ▶ *Focus on education and lifestyle advice* (eg exercise to ↑insulin sensitivity), healthy eating: ↓saturated fats, ↓sugar, ↑starch-carbohydrate, moderate protein. Foods made just for diabetics are not needed. One could regard bariatric surgery as a cure for DM in selected patients. Be prepared to negotiate HbA_{1c} target and review every 3–6 months. Assess *global* vascular risk; start a high-intensity statin (p115), eg atorvastatin as tolerated, control BP (p211). Give foot-care (p212). (Pre-) pregnancy care should be in a multidisciplinary clinic (OHCS p23). Advise informing DVLA and not to drive if hypoglycaemic spells (p159; loss of hypoglycaemia awareness may lead to loss of licence; permanent if HGV).

Type 1 DM Insulin (see BOX 'Using insulin').

Type 2 DM See fig 5.5.

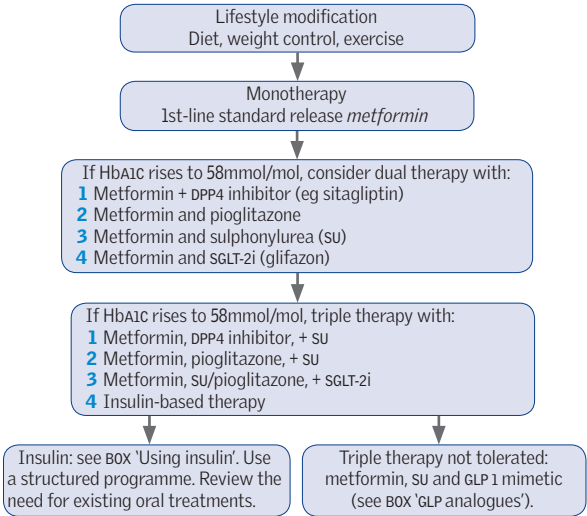


Fig 5.5 Management of type 2 diabetes. Aim for HbA_{1c} 48mmol/mol or 53 if two or more agents. Data from Algorithm for blood glucose lowering therapy in adults with type 2 diabetes, <http://www.nice.org.uk/guidance/ng28/chapter/1-Recommendations#drug-treatment-2>

Oral hypoglycaemic agents

Metformin: A biguanide. ↑ insulin sensitivity and helps weight. SE: nausea; diarrhoea (try modified-release version); abdominal pain; not hypoglycaemia. Avoid if eGFR ≤36mL/min (due to risk lactic acidosis).

DPP4 inhibitors/gliptins: (Eg sitagliptin.) Block the action of DPP-4, an enzyme which destroys the hormone incretin.

Glitazone: ↑ insulin sensitivity; SE: hypoglycaemia, fractures, fluid retention, ↑LFT (do LFT every 8wks for 1yr, stop if ALT up >3-fold). CI: past or present CCF; osteoporosis; monitor weight, and stop if ↑ or oedema.

Sulphonylurea: ↑ insulin secretion; eg gliclazide 40mg/d. SE: hypoglycaemia (monitor glucose); it ↑ weight.

SGLT: Selective sodium-glucose co-transporter-2 inhibitor. Blocks the reabsorption of glucose in the kidneys and promotes excretion of excess glucose in the urine (eg empagliflozin, shown to reduce mortality from cardiovascular disease in patients with type 2 DM, when compared to placebo).⁸

Using insulin

Vital to educate to self-adjust doses in the light of exercise, fingerprick glucose, calorie intake, and carbohydrate counting. •Phone support (trained nurse 7/24). •Can modify diet wisely and avoid binge drinking (danger of delayed hypoglycaemia). •Partner can abort hypoglycaemia: sugary drinks; GlucoGel® PO if coma (no risk of aspiration). •Dose titration to target—eg by 2–4 UNIT steps.

▶▶ It is vital to write UNITS in full when prescribing insulin to avoid misinterpretation of U for zero!

Subcutaneous insulins Short-, medium-, or long-acting. Strength: 100u/mL.

- 1 Ultra-fast acting (Humalog®; Novorapid®); inject at start of meal, or just after (unless sugar-laden)—helps match what is actually eaten (vs what is planned).
- 2 Isophane insulin (variable peak at 4–12h): favoured by NICE (it's cheap!).
- 3 Pre-mixed insulins (eg NovoMix® 30 = 30% short-acting and 70% long-acting).
- 4 Long-acting recombinant human insulin analogues (*insulin glargine*) are used at bedtime in type 1 or 2 DM. There is no awkward peak, so good if nocturnal hypoglycaemia is an issue. Caution if considering pregnancy. *Insulin detemir* is similar and has a role in intensive insulin regimens for overweight type 2 DM.

Common insulin regimens ▶*Plan the regimen to suit the lifestyle, not vice versa.* Disposable pens: dial dose; insert needle 90° to skin. Vary injection site (outer thigh/abdomen); change needle daily.

- 'BD biphasic regimen': twice daily premixed insulins by pen (eg NovoMix 30®)—useful in type 2 DM or type 1 with regular lifestyle.
- 'QDS regimen': before meals ultra-fast insulin + bedtime long-acting analogue: useful in type 1 DM for achieving a flexible lifestyle (eg for adjusting doses with size of meals, or exercise).
- Once-daily before-bed long-acting insulin: a good initial insulin regimen when switching from tablets in type 2 DM. Typical dose to work up to (slowly!): $\geq 1\text{u}/24\text{h}$ for every unit of BMI in adults. Consider retaining metformin ($\pm$ pioglitazone) if needed for tight control and patient is unable to use BD regimen.

Dose adjustment for normal eating (DAFNE): Multidisciplinary teams promoting autonomy can save lives. DAFNE found that training in flexible, intensive insulin dosing improved glycaemic control as well as well-being.⁹ It is resource intensive.

Subcutaneous insulin dosing during intercurrent illnesses (eg influenza)

- ▶▶ Advise patients to avoid stopping insulin during acute illness.
- Illness often increases insulin requirements despite reduced food intake.
 - Maintain calorie intake, eg using milk.
 - Check blood glucose ≥ 4 times a day and look for ketonuria. Increase insulin doses if glucose rising. Advise to get help from a specialist diabetes nurse or GP if concerned (esp. if glucose levels are rising or ketonuria). One option is 2-hourly ultra fast-acting insulin (eg 6–8u) preceded by a fingerprick glucose check.
 - Admit if vomiting, dehydrated, ketotic (▶▶p832), a child, or pregnant.

Insulin pumps (continuous subcutaneous insulin) Consider when attempts to reach HbA1c with multiple daily injections have resulted in disabling hypoglycaemia or person has been unable to achieve target HbA1c despite careful management.

Glucagon-like peptide (GLP) analogues (exenatide, liraglutide)

Work as incretin mimetics. Incretins are gut peptides that work by augmenting insulin release. Given by subcutaneous injection. Patients must have BMI >35 and specific psychological or other medical problems associated with obesity, or have a BMI lower than 35 kg/m^2 , and for whom insulin therapy would have significant occupational implications, or weight loss would benefit other significant obesity-related comorbidities. To continue a GLP 1 mimetic, a person should have a beneficial metabolic response (a reduction of HbA1c by at least 11mmol/mol) and a weight loss of at least 3% of initial body weight in 6 months.



Prospective studies show that good control of hyperglycaemia is key to preventing microvascular complications in type 1 and 2 DM.¹⁰ ► *Find out what problems are being experienced* (eg glycaemic control, morale, erectile dysfunction—p230).

Assess vascular risk BP control (see BOX 'Controlling blood pressure in diabetes') is crucial for preventing macrovascular disease and ↓mortality. Refer to *smoking* cessation services. Check *plasma lipids*.

Look for complications • Check injection sites for infection or lipohypertrophy (fatty change): advise on rotating sites of injection if present.

• **Vascular disease:** Chief cause of death. MI is 4-fold commoner in DM and is more likely to be 'silent'. Stroke is twice as common. Women are at high risk—DM removes the vascular advantage conferred by the female sex. Address other risk factors—diet, smoking, hypertension (p93). Suggest a statin (eg atorvastatin 20mg nocte) for all, even if no overt IHD, vascular disease, or microalbuminuria. Aspirin 75mg reduces vascular events (in context of secondary prevention). Safe to use in diabetic retinopathy.¹¹

• **Nephropathy:** (p314.) Microalbuminuria is when urine dipstick is -ve for protein but the urine albumin:creatinine ratio (UA:CR) is ≥3mg/mmol (units vary, check lab) reflecting early renal disease and ↑vascular risk. If UA:CR >3, inhibiting the renin-angiotensin system with an ACE-i or sartan, even if BP is normal, protects the kidneys. Spironolactone may also help.¹² Refer if UA:CR >7 ± GFR falling by >5mL/min/1.73m²/yr.¹³

• **Diabetic retinopathy:** Blindness is *preventable*. ► *Annual retinal screening mandatory for all patients*. Refer to an ophthalmologist if pre-proliferative changes or if any uncertainty at or near the macula (the only place capable of 6/6 vision). ► Pre-symptomatic screening enables laser photocoagulation to be used, aimed to stop production of angiogenic factors from the ischaemic retina. Indications: maculopathy or proliferative retinopathy. See **figs 5.6–5.9**.

• **Background retinopathy:** Microaneurysms (dots), haemorrhages (blots), and hard exudates (lipid deposits). Refer if near the macula, eg for intravitreal triamcinolone.

• **Pre-proliferative retinopathy:** Cotton-wool spots (eg infarcts), haemorrhages, venous beading. These are signs of retinal ischaemia. Refer to a specialist.

• **Proliferative retinopathy:** New vessels form. Needs urgent referral.

• **Maculopathy:** (Hard to see in early stages.) Suspect if ↓acuity. Prompt laser, intravitreal steroids, or anti-angiogenic agents may be needed in macular oedema. **Pathogenesis:** Capillary endothelial change → vascular leak → microaneurysms → capillary occlusion → local hypoxia + ischaemia → new vessel formation. High retinal blood flow caused by hyperglycaemia (and ↑BP and pregnancy) triggers this, causing capillary pericyte damage. Microvascular occlusion causes *cotton-wool spots* (± *blot haemorrhages* at interfaces with perfused retina). *New vessels* form on the disc or ischaemic areas, proliferate, bleed, fibrose, and can detach the retina. Aspirin² (2mg/kg/d) may be recommended by ophthalmologists; there is no evidence that it ↑ bleeding.

• **Cataracts:** May be juvenile 'snowflake' form, or 'senile'—which occur earlier in diabetic subjects. Osmotic changes in the lens induced in acute hyperglycaemia reverse with normoglycaemia (so wait before buying glasses).

• **Rubeosis iridis:** New vessels on iris: occurs late and may lead to glaucoma.

• **Metabolic complications:** p832.

• **Diabetic feet:** p212.

• **Neuropathy:** p212.

² As DM has so many vascular events, particularly encourage statin use (p690), esp. if LDL >3mmol/L or systolic BP >140. Even consider a statin *whatever* the pre-treatment cholesterol; discuss with your patient.

Controlling blood pressure in diabetes

Type 1 DM: Treat BP if $>135/85\text{mmHg}$, unless albuminuria or two or more features of metabolic syndrome, in which case it should be $130/80\text{mmHg}$ (NICE 2015). Use an ACE-i 1st line or angiotension receptor antagonist if intolerant. If hypertensive and underlying renal involvement, see local guidance, p304.

Type 2 DM: Target BP $<140/80\text{mmHg}$ or $<130/80\text{mmHg}$ if kidney, eye, or cerebrovascular damage. 1st-line drug treatment should be an ACE-i, except in those of African or Caribbean origin, where ACE-i plus diuretic or a calcium-channel antagonist (CCA) should be started. For pregnant women offer CCA. Do not offer aspirin for the primary prevention of cardiovascular disease to adults with type 1 DM (NICE 2015). Don't combine an ACE-i with an angiotension receptor antagonist.



Fig 5.6 Background retinopathy, with microaneurysms and hard exudates.

Courtesy of Prof J Trobe.

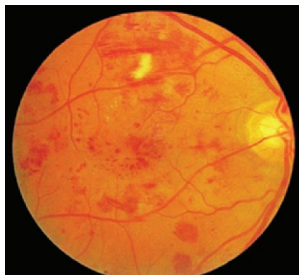


Fig 5.7 Pre-proliferative retinopathy, with haemorrhages and a cotton-wool spot.

Reproduced from Warrell *et al*, *Oxford Textbook of Medicine*, 2010, with permission from Oxford University Press.



Fig 5.8 Proliferative retinopathy, with new vessel formation and haemorrhages.

Courtesy of Prof J Trobe.



Fig 5.9 Scars from previous laser photocoagulation.

Courtesy of Prof J Trobe.

Improving quality of life: going beyond the pleasures of the flesh

'I cannot eat what I want because of your pitiful diet. Sex is out because diabetes has made me impotent. Smoking is banned, so what's left? I'd shoot myself if only I could see straight.' Start by acknowledging your patient's distress. Don't shrug it off—but don't take it at face value either. Life may be transformed by cataract surgery, sildenafil (unless contraindicated, p230), dietary negotiation, and sport (it needn't be shooting). Take steps to simplify care. Stop blood glucose self-monitoring if it's achieving nothing (known to ↓ quality of life). Even if all these interventions fail, you have one trump card up your sleeve: 'Let's both try to find one new thing of value before we next meet—and compare notes'. This opens the way to vicarious pleasure: a whole new world.

► Refer *early* to foot services (podiatry, imaging, vascular surgery).

Amputations are common (135/week)—and preventable: *good care saves legs*. Examine feet regularly. Distinguish between ischaemia (critical toes ± absent foot pulses and worse outcome) and peripheral neuropathy (injury or infection over pressure points, eg the metatarsal heads). In practice, many have both.

Neuropathy ↓ Sensation in 'stocking' distribution: test sensation with a 10g monofilament fibre (sensory loss is patchy so examine all areas), absent ankle jerks, neuropathic deformity (Charcot joint, **fig 5.11**): pes cavus, claw toes, loss of transverse arch, rocker-bottom sole. Caused by loss of pain sensation, leading to mechanical stress and repeated joint injury. Swelling, instability, and deformity. ► Early recognition is vital (cellulitis or osteomyelitis are often misdiagnosed).

Ischaemia If the foot pulses cannot be felt, do Doppler pressure measurements. Any evidence of neuropathy or vascular disease raises risk of foot ulceration. *Educate* (daily foot inspection—eg with a mirror for the sole; comfortable shoes). *Regular chiropody* to remove callus, as haemorrhage and tissue necrosis may occur below, leading to ulceration. *Treat fungal infections* (p408). *Surgery* (including endovascular angioplasty balloons, stents, and subintimal recanalization) has a role.

Foot ulceration Typically painless, punched-out ulcer (**fig 5.10**) in an area of thick callus ± superadded infection. Causes cellulitis, abscess ± osteomyelitis.

Assess degree of:

1 Neuropathy (clinically).

2 Ischaemia (clinically + Doppler ± angiography).

3 Bony deformity, eg Charcot joint (clinically + x-ray). See **fig 5.11**.

4 Infection (swabs, blood culture, x-ray for osteomyelitis, probe ulcer to reveal depth).

Management: Regular chiropody. Bed rest ± therapeutic shoes. For Charcot joints: bed rest/crutches/total contact cast until oedema and local warmth reduce and bony repair is complete (≈8wks). Bisphosphonates may help. Charcot joints are also seen in tabes dorsalis, spina bifida, syringomyelia, and leprosy. Metatarsal head surgery may be needed. If there is cellulitis, admit for IV antibiotics. Common organisms: staphs, streps, anaerobes. Start empirically (as per your local guidance) with benzylpenicillin 1.2g/6h IV and flucloxacillin 1g/6h IV ± metronidazole 500mg/8h IV. IV insulin may improve healing. Get surgical help early. The degree of peripheral vascular disease, general health, and patient request will determine degree of vascular reconstruction/surgery.

► **Absolute indications for surgery:** Abscess or deep infection; spreading anaerobic infection; gangrene/rest pain; suppurative arthritis.

Diabetic neuropathies *Symmetric sensory polyneuropathy:* ('glove & stocking' numbness, tingling, and pain, eg worse at night). *R:* paracetamol → tricyclic (amitriptyline 10–25mg nocte; gradually ↑ to 150mg) → duloxetine, gabapentin, or pregabalin → opiates. Avoiding weight-bearing helps. *Mononeuritis multiplex:* (eg III & VI cranial nerves). Treatment: hard! If sudden or severe, immunosuppression may help (corticosteroids, IV immunoglobulin, ciclosporin). *Amyotrophy:* Painful wasting of quadriceps and other pelfemoral muscles. Use electrophysiology to show, eg lumbosacral radiculopathy, plexopathy, or proximal crural neuropathy. Natural course: variable with gradual but often incomplete improvement. IV immunoglobulins have been used. *Autonomic neuropathy:* (p505) Postural BP drop; ↓cerebrovascular autoregulation; loss of respiratory sinus arrhythmia (vagal neuropathy); gastroparesis; urine retention; erectile dysfunction; gustatory sweating; diarrhoea (may respond to codeine phosphate). Gastroparesis (early satiety, post-prandial bloating, nausea/vomiting) is diagnosed by gastric scintigraphy with a ⁹⁹technetium-labelled meal; anti-emetics, erythromycin, or gastric pacing. Postural hypotension may respond to fludrocortisone (SE: oedema, ↑BP)/midodrine (α-agonist; SE: ↑BP).



Preventing loss of limbs: primary or secondary prevention?

Traditionally prevention involves foot care advice in diabetic clinics (eg 'Don't go bare-foot'), promoting euglycaemia and normotension. But despite this, the sight of a diabetic patient minus one limb is not rare, and must prompt us to redouble our commitment to primary prevention, ie stopping those at risk from ever getting diabetes. The sequelae of diabetic neuropathy can lead to gangrene, amputation, and the impact on quality of life can be profound. As one patient post amputation said, 'I begin again to walk, on crutches. What nuisance, what fatigue, what sadness, when I think about all my ancient travels, and how active I was just 5 months ago! Where are the runnings across mountains, the walks, the deserts, the rivers, and the seas? And now, the life of a legless cripple. For I begin to understand that crutches, wooden and articulated legs, are a pack of jokes...Goodbye to family, goodbye to future! My life is gone, I'm no more than an immobile trunk' (Arthur Rimbaud. Letter to his sister Isabelle, 10 July 1891).



Fig 5.10 Gangrene (toes 2, 4, and 5).

Reproduced from Warrell *et al*, *Oxford Textbook of Medicine*, 2010, with permission from Oxford University Press.



Fig 5.11 Charcot (neuropathic) joint.

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Special situations in diabetes

Pregnancy: (OHCS p23) 4% are complicated by DM: either pre-existing (<0.5%), or new-onset gestational diabetes (GDM) (>3.5%).

- All forms carry an increased risk to mother and fetus: miscarriage, pre-term labour, pre-eclampsia, congenital malformations, macrosomia, and a worsening of diabetic complications, eg retinopathy, nephropathy.
- Risk of GDM ↑ if: aged over 25; family history; +ve; tweight; non-Caucasian; HIV+ve; previous gestational DM.
- Pre-conception: offer general advice, and discuss risks. Control/reduce weight, aim for good glucose control, offer folic acid 5mg/d until 12 weeks.
- Screen for GDM with OGTT if risk factors at booking (16–18 weeks if previous GDM).
- Oral hypoglycaemics other than metformin should be discontinued. Metformin may be used as an adjunct or alternative to insulin in type 2 DM or GDM.

6wks postpartum, do a fasting glucose. Even if –ve, 50% go on to develop DM.

Surgery: Optimal blood sugar control pre, peri, and post operatively is important to minimize risk of infection and balance catabolic response to surgery. Type 1 diabetics should ideally be first on the list and BMs should have been stabilized 1–2 days pre major surgery. Consult local policy for how to manage insulin-treated/non-insulin-treated patients on morning of surgery (eg setting up glucose/insulin infusion).

Acute illness: Diabetics are prone to hyperglycaemia during periods of illness, in spite of reduced oral intake. Avoid stopping insulin in periods of acute illness.

►► **Hypoglycaemia** Commonest endocrine emergency—see p834. Prompt diagnosis and treatment essential—brain damage & death can occur if severe or prolonged.

Definition Plasma glucose $\leq 3\text{mmol/L}$. Threshold for symptoms varies. See BOX.

Symptoms *Autonomic*: Sweating, anxiety, hunger, tremor, palpitations, dizziness.

Neuroglycopenic: Confusion, drowsiness, visual trouble, seizures, coma. Rarely focal symptoms, eg transient hemiplegia. Mutism, personality change, restlessness, and incoherence may lead to misdiagnosis of alcohol intoxication or even psychosis.

Fasting hypoglycaemia *Causes*: The chief cause is insulin or sulfonylurea treatment in a diabetic, eg $\uparrow$ activity, missed meal, accidental or non-accidental overdose (check for circulating oral hypoglycaemics). In *non-diabetics* you must **EXPLAIN** mechanism: **E**xogenous drugs, eg insulin, oral hypoglycaemics (p208)? access through diabetic in the family? Body-builders may misuse insulin to help stamina. Also: alcohol, eg a binge with no food; aspirin poisoning; ACE-i; β -blockers; pentamidine; quinine sulfate; aminoglutethamide; insulin-like growth factor.¹⁴

Pituitary insufficiency.

Liver failure, plus some rare inherited enzyme defects.

Addison's disease.

Islet cell tumours (insulinoma) and immune hypoglycaemia (eg anti-insulin receptor antibodies in Hodgkin's disease).

Non-pancreatic neoplasms, eg fibrosarcomas and haemangiopericytomas.

When to investigate:

- Whipple answered this (*Whipple's triad*): symptoms or signs of hypoglycaemia + $\downarrow$ plasma glucose + resolution of symptoms or signs post glucose rise.
- Document BM during attack and lab glucose if in hospital (monitors often not reliable at low readings).
- Take a drug history and exclude liver failure.
- 72h fasting may be needed (monitor closely). Bloods: glucose, insulin, c-peptide, and plasma ketones if symptomatic. If endogenous hyperinsulinism suspected, do insulin, c-peptide, proinsulin, β -hydroxybutyrate.

Interpreting results:

- Hypoglycaemic hyperinsulinaemia (HH): *Causes*: insulinoma, sulfonylureas, insulin injection (no detectable c-peptide—only released with endogenous insulin); non-insulinoma pancreatogenous hypoglycaemia syndrome, mutation in the insulin-receptor gene. Congenital HH follows mutations in genes involved in insulin secretion (ABCC8, KCNJ11, GLUD1, CGK, HADH, SLC16A1, HNF4A, ABCC8, & KCNJ11).¹⁵
- Insulin low or undetectable, no excess ketones. *Causes*: non-pancreatic neoplasm; anti-insulin receptor antibodies.
- $\downarrow$ Insulin, $\uparrow$ ketones. *Causes*: alcohol, pituitary insufficiency, Addison's disease.

Post-prandial hypoglycaemia May occur after gastric/bariatric surgery ('dumping', p623), and in type 2 DM. *Investigation*: Prolonged OGTT (5h, p206).

Treatment ►► See p834. If episodes are often, advise many small high-starch meals. If post-prandial $\downarrow$ glucose, give slowly absorbed carbohydrate (high fibre). In diabetics, rationalize insulin therapy (p209).

The definition of hypoglycaemia is context-dependent

The brain stops working if plasma glucose levels get too low, so we are nervous of levels $\leq 3\text{mmol/L}$. But some are asymptomatic at this level. So what is definitely abnormal? The answer may be 4mmol/L , allowing for inaccuracies in fingerprick BMs (NB: whole blood glucose is 10–15% < plasma glucose.) Think: '*In this ill patient when can I be sure that a low glucose is not contributing to their illness?*' If $<4\text{mmol/L}$, you may be wise to treat (p834)—just in case. Consider, is the patient on hypoglycaemics, have they binged on alcohol 24hrs pre test? Skipped meals? Is there an underlying illness, eg insulinoma? Unlikely, but possible. Keep an open mind; let the GP know. Counsel patient and relative about warning signs of hypoglycaemia. Be more inclined to investigate if the effects of even mild hypoglycaemia might be disastrous (eg in pilots) or if there are unexplained symptoms.

Insulinoma

This often benign (90–95%) pancreatic islet cell tumour is sporadic or seen with MEN-1 (p223). It presents as fasting hypoglycaemia, with Whipple's triad:

- 1 Symptoms associated with fasting or exercise.
- 2 Recorded hypoglycaemia with symptoms.
- 3 Symptoms relieved with glucose.

Screening test Hypoglycaemia + ↑plasma insulin during a long fast.

Suppressive tests Give IV insulin and measure C-peptide. Normally exogenous insulin suppresses C-peptide production, but this does not occur in insulinoma.

Imaging CT/MRI ± endoscopic pancreatic US ± IACS (see Box; all fallible, so don't waste too much time before proceeding to intra-operative visualization ± intra-operative ultrasound). ¹⁸F-L-3,4-dihydroxyphenylalanine PET-CT can help guide laparoscopic surgery.

Treatment Excision.

Nesidioblastosis See Box. If this doesn't work, options are: diet, diazoxide, dextrose IV, enteral feeding, everolimus.

▶ **Diabetic ketoacidosis** Results from insulin deficiency (eg unknown diagnosis, intercurrent illness, interruption of insulin therapy). See p832.

Pursuing a voyage to the islets of Langerhans to the bitter end

A 50-year-old had episodic early-morning sweats and tremors and was found to have hyperinsulinaemic hypoglycaemia (= Nesidioblastosis). Selective intra-arterial calcium infusions (IACS) showed a 2-fold increase in insulin secretion after infusion of the splenic and superior mesenteric arteries, so setting the stage for 'hunt the insulinoma'.

But cross-sectional imaging and endoscopic ultrasound were normal. At laparotomy, no lesion was found despite mobilization of the pancreas, or during intra-operative ultrasound. 'Time to sew up and go home?' 'No!' said the surgeon, 'I'm going to do a distal pancreatectomy'. Histology showed no discrete insulinoma, but diffuse islet cell hyperplasia (*nesidioblastosis*). How much pancreas to resect? Too little and nothing is gained: too much spells pancreatic endocrine disaster. Luckily the surgeon guessed right, and the patient was cured by the procedure.¹⁶



Physiology

Thyroid-stimulating hormone (TSH=thyrotropin), a glycoprotein, is produced from the anterior pituitary (fig 5.12). The thyroid produces mainly T_4 , which is 5-fold less active than T_3 . 85% of T_3 is formed from peripheral conversion of T_4 . Most T_3 and T_4 in plasma is protein bound, eg to thyroxine-binding globulin (TBG). The unbound portion is the active part. T_3 and T_4 $\uparrow$ cell metabolism, via nuclear receptors, and are thus vital for growth and mental development. They also $\uparrow$ catecholamine effects. Thyroid hormone abnormalities are usually due to problems in the thyroid gland itself, and rarely caused by the hypothalamus or the anterior pituitary.

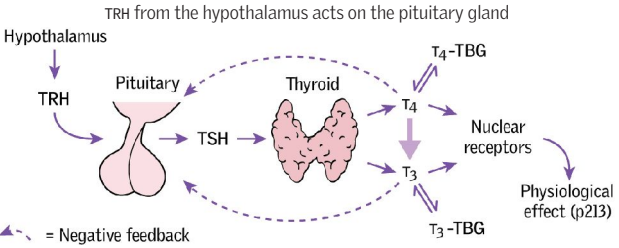


Fig 5.12 Pathways involved in thyroid function.

Basic tests See table 5.2. Free T_4 and T_3 are more useful than total T_4 and T_3 as the latter are affected by TBG. Total T_4 and T_3 are $\uparrow$ when TBG is $\uparrow$ and *vice versa*. TBG is $\uparrow$ in pregnancy, oestrogen therapy (HRT, oral contraceptives), and hepatitis. TBG is $\downarrow$ in nephrotic syndrome and malnutrition (both from protein loss), drugs (androgens, corticosteroids, phenytoin), chronic liver disease, and acromegaly. TSH is very useful:

- **Hyperthyroidism suspected:** Ask for T_3 , T_4 , and TSH. All will have $\downarrow$ TSH (except the rare TSH-secreting pituitary adenoma). Most have $\uparrow$ T_4 , but $\sim 1\%$ have only raised T_3 .
- **Hypothyroidism suspected or monitoring replacement R_x :** Ask for only T_4 and TSH. T_3 does not add any extra information. TSH varies through the day: trough at 2PM; 30% higher during darkness, so during monitoring, try to do at the same time.

Sick euthyroidism: In any systemic illness, TFTs may become deranged. The typical pattern is for 'everything to be low'. The test should be repeated after recovery.

Assay interference is caused by antibodies in the serum, interfering with the test.

Other tests

- **Thyroid autoantibodies:** Antithyroid peroxidase (TPO; formerly called microsomal) antibodies or antithyroglobulin antibodies may be increased in autoimmune thyroid disease: Hashimoto's or Graves' disease. If +ve in Graves', there is an increased risk of developing hypothyroidism at a later stage.
- **TSH receptor antibody:** May be $\uparrow$ in Graves' disease (useful in pregnancy).
- **Serum thyroglobulin:** Useful in monitoring the treatment of carcinoma (p600), and in detection of factitious (self-medicated) hyperthyroidism, where it is low.
- **Ultrasound:** This distinguishes cystic (usually, but not always, benign) from solid (possibly malignant) nodules. If a solitary (or dominant) large nodule, in a multinodular goitre, do a fine-needle aspiration to look for thyroid cancer; see fig 13.23, p601.
- **Isotope scan:** (123 Iodine, 99m technetium pertechnetate, etc; see fig 13.22, p601.) Useful for determining the cause of hyperthyroidism and to detect retrosternal goitre, ectopic thyroid tissue or thyroid metastases (+ whole body CT). If there are suspicious nodules, the question is: does the area have increased (hot), decreased (cold), or the same (neutral) uptake of isotope as the remaining thyroid (see fig 5.13). Few neutral and almost no hot nodules are malignant. 20% of 'cold' nodules are malignant. Surgery is most likely to be needed if:
 - rapid growth
 - compression signs
 - dominant nodule on scintigraphy
 - nodule ≥ 3 cm
 - hypo-echogenicity. See also p738.

Screen the following for abnormalities in thyroid function

- Patients with atrial fibrillation.
- Patients with hyperlipidaemia (4–14% have hypothyroidism).
- Diabetes mellitus—on annual review.
- Women with type 1 DM during 1st trimester and post delivery (3-fold rise in incidence of postpartum thyroid dysfunction).
- Patients on amiodarone or lithium (6 monthly).
- Patients with Down's or Turner's syndrome, or Addison's disease (yearly).

Table 5.2 Interpreting TFTs

Hormone profile	Diagnosis
↑TSH, ↓T ₄	Hypothyroidism
↑TSH, normal T ₄	Treated hypothyroidism or subclinical hypothyroidism (p221)
↑TSH, ↑T ₄	TSH-secreting tumour or thyroid hormone resistance
↑TSH, ↑T ₄ and ↓T ₃	Slow conversion of T ₄ to T ₃ (deiodinase deficiency; euthyroid hyperthyroxinaemia) or thyroid hormone antibody artefact
↓TSH, ↑T ₄ or ↑T ₃	Hyperthyroidism
↓TSH, normal T ₄ and T ₃	Subclinical hyperthyroidism
↓TSH, ↓T ₄	Central hypothyroidism (hypothalamic or pituitary disorder)
↓TSH, ↓T ₄ and ↓T ₃	Sick euthyroidism or pituitary disease
Normal TSH, abnormal T ₄	Consider changes in thyroid-binding globulin, assay interference, amiodarone, or pituitary TSH tumour

*In 'consumptive hypothyroidism' deiodinase activity is ↑↑; suspect if thyroxine doses have to be ↑↑.

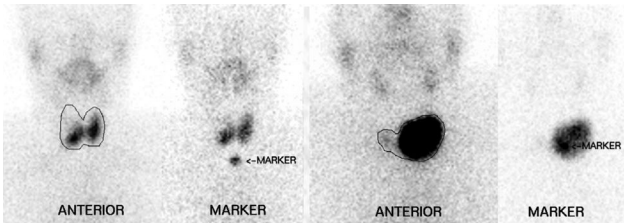


Fig 5.13 The images are from an isotope scan, with and without markers placed over the sternal notch. We can see on the left that the nodule is metabolically inactive ('cold'). The hot nodule (right pair) is a very avid nodule causing background thyroid suppression.

Image courtesy of Dr Y. T. Huang.



The clinical effect of excess thyroid hormone, usually from gland hyperfunction.

Symptoms Diarrhoea; ↓weight; ↑appetite (if ↑↑, paradoxical weight *gain* in 10%); over-active; sweats; heat intolerance; palpitations; tremor; irritability; labile emotions; oligomenorrhoea ± infertility. Rarely psychosis; chorea; panic; itch; alopecia; urticaria.

Signs Pulse fast/irregular (AF or SVT; VT rare); warm moist skin; fine tremor; palmar erythema; thin hair; lid lag; lid retraction (exposure of sclera above iris; causing 'stare', (fig 5.14); eyelid lags behind eye's descent as patient watches your finger descend slowly). There may be goitre (fig 5.15); thyroid nodules; or bruit depending on the cause. **Signs of Graves' disease:** 1 *Eye disease* (BOX 'Thyroid eye disease'): exophthalmos, ophthalmoplegia. 2 *Pretibial myxoedema*: oedematous swellings above lateral malleoli: the term *myxoedema* is confusing here. 3 *Thyroid acropachy*: extreme manifestation, with clubbing, painful finger and toe swelling, and periosteal reaction in limb bones.

Tests ↓TSH (suppressed), T_4 , and ↑ T_3 . There may be mild normocytic anaemia, mild neutropenia (in Graves'), ↑ESR, ↑ Ca^{2+} , ↑LFT. **Also:** Check thyroid autoantibodies. Isotope scan if the cause is unclear, to detect nodular disease or subacute thyroiditis. If ophthalmopathy, test visual fields, acuity, and eye movements (see BOX 'Thyroid eye disease').

Causes

Graves' disease: Prevalence: 0.5% (% of cases of hyperthyroidism). ♀:♂≈9:1. Typical age: 40–60yrs (younger if maternal family history). Cause: circulating IgG autoantibodies binding to and activating G-protein-coupled thyrotropin receptors, which cause smooth thyroid enlargement and hormone production (esp. T_3), and react with orbital autoantigens. Triggers: stress; infection; childbirth. Patients are often hyperthyroid but may be, or become, hypo- or euthyroid. It is associated with other autoimmune diseases: vitiligo, type 1 DM, Addison's (table 5.3).

Toxic multinodular goitre: Seen in the elderly and in iodine-deficient areas. There are nodules that secrete thyroid hormones. Surgery is indicated for compressive symptoms from the enlarged thyroid (dysphagia or dyspnoea).

Toxic adenoma: There is a solitary nodule producing T_3 and T_4 . On isotope scan, the nodule is 'hot' (p216), and the rest of the gland is suppressed.

Ectopic thyroid tissue: Metastatic follicular thyroid cancer, or struma ovarii: ovarian teratoma with thyroid tissue.

Exogenous: Iodine excess, eg food contamination, contrast media (→thyroid storm, p834, if already hyperthyroid). Levothyroxine excess causes ↑ T_4 , ↑ T_3 , ↑thyroglobulin.

Others: 1 *Subacute de Quervain's thyroiditis*: self-limiting post-viral with painful goitre, ↑↑° ± ↑ESR. Low isotope uptake on scan. **R:** NSAIDs. 2 *Drugs*: amiodarone (p220), lithium (hypothyroidism more common). 3 *Postpartum*. 4 *TB* (rare).

Treatment

- 1 Drugs:** β-blockers (eg propranolol 40mg/6h) for rapid control of symptoms. *Anti-thyroid medication*: two strategies (equally effective):¹⁷ **A Titration**, eg carbimazole 20–40mg/24h PO for 4wks, reduce according to TFTs every 1–2 months. **B Block-replace**: Give carbimazole + levothyroxine simultaneously (less risk of iatrogenic hypothyroidism). In Graves', maintain on either regimen for 12–18 months then withdraw. ~50% will relapse, requiring radioiodine or surgery. Carbimazole SE: agranulocytosis (↓↓neutrophils, can lead to dangerous sepsis; rare (0.03%)); warn to stop and get an urgent FBC if signs of infection, eg ↑ T° t, sore throat/mouth ulcers.
- 2 Radioiodine (^{131}I):** Most become hypothyroid post-treatment. There is no evidence for cancer, birth defects, or infertility in women. CI: pregnancy, lactation. Caution in active hyperthyroidism as risk of thyroid storm (p835).
- 3 Thyroidectomy (usually total):** Carries a risk of damage to recurrent laryngeal nerve (hoarse voice) and hypoparathyroidism. Patients will become hypothyroid, so thyroid replacement needed.
- 4 In pregnancy and infancy:** Get expert help. See OHCS p24 & OHCS p182.

Complications Heart failure (thyrotoxic cardiomyopathy, ↑ in elderly), angina, AF (seen in 10–25%: control hyperthyroidism and warfarinize if no contraindication), osteoporosis, ophthalmopathy, gynaecomastia. →Thyroid storm (p835).

Thyroid eye disease

Seen in 25–50% of people with Graves' disease. The main known risk factor is smoking. The eye disease may not correlate with thyroid disease and the patient can be euthyroid, hypothyroid, or hyperthyroid at presentation. Eye disease may be the first presenting sign of Graves' disease, and can also be worsened by treatment, typically with radioiodine (usually a transient effect). Retro-orbital inflammation and lymphocyte infiltration results in swelling of the orbit.

Symptoms Eye discomfort, grittiness, ↑tear production, photophobia, diplopia, ↓acuity, afferent pupillary defect (p72) may mean optic nerve compression: ▶*Seek expert advice at once as decompression may be needed.* Nerve damage does not necessarily go hand-in-hand with protrusion. Indeed, if the eye cannot protrude for anatomical reasons, optic nerve compression is more likely—a paradox!

Signs Exophthalmos—appearance of protruding eye; proptosis—eyes protrude beyond the orbit (look from above in the same plane as the forehead); conjunctival oedema; corneal ulceration; papilloedema; loss of colour vision. Ophthalmoplegia (especially of upward gaze) occurs due to muscle swelling and fibrosis.

Tests Diagnosis is clinical. CT/MRI of the orbits may reveal enlarged eye muscles.

Management Get specialist help. Treat hyper- or hypothyroidism. Advise to stop smoking (worse prognosis). Most have mild disease that can be treated symptomatically (artificial tears, sunglasses, avoid dust, elevate bed when sleeping to ↓periorbital oedema). Diplopia may be managed with a Fresnel prism stuck to one lens of a spectacle (aids easy changing as the exophthalmos changes). In more severe disease, try high-dose steroids (iv methylprednisolone is better than prednisolone 100mg/day po)¹⁸—decreasing according to symptoms. Surgical decompression is used for severe sight-threatening disease, or for cosmetic reasons once the activity of eye disease has reduced (via an inferior orbital approach, using space in the ethmoidal, sphenoidal, and maxillary sinuses). Eyelid surgery may improve cosmesis and function. Orbital radiotherapy can be used to treat ophthalmoplegia but has little effect on proptosis. *Future options:* Anti-TNFα antibodies (eg infliximab).



Fig 5.14 Thyroid eye disease: lid retraction causing a 'staring' appearance.

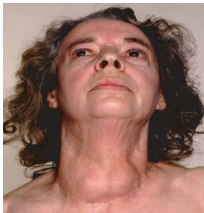


Fig 5.15 Goitre.

Causes of goitre

Diffuse

- Physiological
- Graves' disease
- Hashimoto's thyroiditis
- Subacute (de Quervain's) thyroiditis (painful).

Nodular

- Multinodular goitre
- Adenoma
- Carcinoma.

Table 5.3 Manifestations of Graves' disease—and pathophysiology

Pituitary	Suppressed TSH	↓Expression of thyrotropin β subunit
Heart	↑Rate; ↑contractility	↑Serum atrial natriuretic peptide
Liver	↑Peripheral T ₃ ; LDL↓ (p690)	↑Type 1 5'-deiodinase; LDL receptors
Bone	↑Bone turnover; osteoporosis	↑Osteocalcin; ↑ALP; ↑urinary N-telopeptide
Genital ♂	↓Libido; erectile dysfunction	↑Sex hormone globulin; ↓testosterone
Genital ♀	Irregular menses	Oestrogen antagonism
Metabolic	↑Thermogenesis; ↑O ₂ use	↑Fatty acid oxidation; ↑Na-K ATPase
White fat	↓Fat mass	↑Adrenergic-mediated lipolysis
CNS	Stiff person syndrome (rare)*	Antibodies to glutamic acid decarboxylase
Muscle	Proximal myopathy	↑Sarcoplasmic reticulum Ca ²⁺ -activated ATPase
Thyroid	↑Secretion of T ₃ and T ₄	↑Type 2 5'-deiodinase activity in thyroid

*Emotional or tactile stimuli cause spasms; seen in any autoimmune state (eg type 1 DM); R₁: baclofen± IV Ig.

The clinical effect of lack of thyroid hormone. It is common (4/1000/yr). If treated, prognosis is excellent; untreated it is disastrous (eg heart disease, dementia). ► *As it is insidious, both you and your patient may not realize anything is wrong*, so be alert to subtle, non-specific symptoms, esp. in women ≥ 40 yrs old ($\sigma:\sigma \approx 6:1$).

Symptoms Tiredness; sleepy, lethargic; $\downarrow$ mood; cold-disliking; $\uparrow$ weight; constipation; menorrhagia; hoarse voice; $\downarrow$ memory/cognition; dementia; myalgia; cramps; weakness.

Signs **BRADYCARDIC**; reflexes relax slowly; ataxia (cerebellar); dry thin hair/skin; yawning/drowsy/coma (p834); cold hands $\pm \downarrow T^\circ$; ascites $\pm$ non-pitting oedema (lids; hands; feet) $\pm$ pericardial or pleural effusion; round puffy face/double chin/obese; defeated demeanour; immobile $\pm$ ileus; ccf. Also: neuropathy; myopathy; goitre (fig 5.16).

Diagnosis (p216) ► Have a low threshold for doing TFTs! $\uparrow$ TSH (eg ≥ 4 mU/L);³ $\downarrow T_4$ (in rare secondary hypothyroidism: $\downarrow T_4$ and $\downarrow$ TSH or $\leftrightarrow$ due to lack from the pituitary, p232). Cholesterol and triglycerides; macrocytosis (less often normochromic anaemia too).

Causes of primary autoimmune hypothyroidism

- **Primary atrophic hypothyroidism**: $\sigma:\sigma \approx 6:1$. Common. Diffuse lymphocytic infiltration of the thyroid, leading to atrophy, hence no goitre.
- **Hashimoto's thyroiditis**: Goitre due to lymphocytic and plasma cell infiltration. Commoner in women aged 60–70 yrs. May be hypothyroid or euthyroid; rarely initial period of hyperthyroid ('Hashitoxicosis'). Autoantibody titres are very high.

Other causes of primary hypothyroidism

- World-wide the chief cause is **iodine deficiency**.
- **Post-thyroidectomy or radioiodine treatment**.
- **Drug-induced**: Antithyroid drugs, amiodarone, lithium, iodine.
- **Subacute thyroiditis**: Temporary hypothyroidism after hyperthyroid phase.

Secondary hypothyroidism Not enough TSH (due to hypopituitarism); very rare.

Hypothyroidism's associations Autoimmune is seen with other autoimmune diseases (type 1 DM, Addison's, and PA, p334). Turner's and Down's syndromes, cystic fibrosis, primary biliary cholangitis, ovarian hyperstimulation (OHCS p311); POEMS syndrome—polyneuropathy, organomegaly, endocrinopathy, m-protein band (plasmacytoma) + skin pigmentation/tethering. **Genetic**: Dysmorphogenesis: genetic (often autosomal recessive) defect in hormone synthesis, eg Pendred's syndrome (with deafness): there is uptake on isotope scan, which is displaced by potassium perchlorate.

Pregnancy problems Eclampsia, anaemia, prematurity, $\downarrow$ birthweight, stillbirth, PPH.

Treatment

- **Healthy and young**: Levothyroxine (T_4), 0–100 mcg/24h po; review at 12 wks. Adjust 6-weekly by clinical state and to normalize but not suppress TSH (keep TSH > 0.5 mU/L). Thyroxine's $t_{1/2}$ is ~ 7 d, so wait ~ 4 wks before checking TSH to see if a dose change is right. NB: small changes in serum free T_4 have a logarithmic effect on TSH. Once normal, check TSH yearly. Enzyme inducers (p689) $\uparrow$ metabolism of levothyroxine.
- **Elderly or ischaemic heart disease**: Start with 25 mcg/24h; $\uparrow$ dose by 25 mcg/4 wks according to TSH (► cautiously, as levothyroxine may precipitate angina or MI).
- **If diagnosis is in question and T_4 already given**: Stop T_4 ; recheck TSH in 6 weeks.

Amiodarone An iodine-rich drug structurally like T_4 ; 2% of users will get significant thyroid problems from it. Hypothyroidism can be caused by toxicity from iodine excess (T_4 release is inhibited). Thyrotoxicosis may be caused by a destructive thyroiditis causing hormone release. Here, radioiodine uptake can be undetectable and if this is the case, glucocorticoids may help. Get expert help. Thyroidectomy may be needed if amiodarone cannot be discontinued. $7\frac{1}{2}$ of amiodarone ≈ 80 d, so problems persist after withdrawal. If on amiodarone, check TFTs 6-monthly.

►► **Myxoedema coma** The ultimate hypothyroid state before death. See p834.

3 ► *Treat the patient not the blood level!* No exact cut-off in TSH can be given partly because risk of death from heart disease mirrors TSH even when in the normal range in women. Risk ≈ 1.4 if TSH 1.5–2.4 vs 1.7 if TSH 2.5–3.5. If TSH > 3.65 and possibly symptomatic, a low dose of levothyroxine may be tried. Monitor symptoms, TSH and T_4 carefully. Over-exposure to thyroxine may cause osteoporosis $\pm$ AF.

Why are symptoms of thyroid disease so various, and so subtle?

Almost all our cell nuclei have receptors showing a high affinity for T_3 : that known as $TR\alpha-1$ is abundant in muscle and fat; $TR\alpha-2$ is abundant in brain; and $TR\beta-1$ is abundant in brain, liver, and kidney. These receptors, via their influence on various enzymes, affect the following processes:

- The metabolism of substrates, vitamins, and minerals.
- Modulation of all other hormones and their target-tissue responses.
- Stimulation of O_2 consumption and generation of metabolic heat.
- Regulation of protein synthesis, and carbohydrate and lipid metabolism.
- Stimulation of demand for co-enzymes and related vitamins.

Subclinical thyroid disease

Subclinical hypothyroidism Suspect if $TSH > 4 \text{ mU/L}$ with normal T_4 and T_3 , and no symptoms. It is common: $\sim 10\%$ of those > 55 yrs have $\uparrow TSH$. Risk of progression to frank hypothyroidism is $\sim 2\%$, and increases as $\uparrow TSH$; risk doubles if thyroid peroxidase antibodies are present, and is also increased in men. *Management:*

- Confirm that raised TSH is persistent (recheck in 2–4 months).
- Recheck the history: if any non-specific features (eg depression), discuss benefits of treating (p220) with the patient—maybe they will function better.
- Have a low threshold for carefully supervised treatment as your patient may not be so asymptomatic after all, and cardiac deaths *may* be prevented. Treat if: **1** $TSH \geq 10 \text{ mU/L}$. **2** +ve thyroid autoantibodies. **3** Past (treated) Graves'. **4** Other organ-specific autoimmunity (type 1 DM, myasthenia, pernicious anaemia, vitiligo), as they are more likely to progress to clinical hypothyroidism. *If TSH 4–10, and vague symptoms*, treat for 6 months—only continue if symptoms improve (or the patient is trying to conceive). If the patient does not fall into any of these categories, monitor TSH yearly.
- Risks from well-monitored treatment of subclinical hypothyroidism are small (but there is an $\uparrow$ risk of atrial fibrillation and osteoporosis if over-treated).

Subclinical hyperthyroidism occurs when $\downarrow TSH$, with normal T_4 and T_3 . There is a 41% increase in relative mortality from all causes versus euthyroid control subjects—eg from AF and osteoporosis. *Management:*

- Confirm that suppressed TSH is persistent (recheck in 2–4 months).
- Check for a non-thyroidal cause: illness, pregnancy, pituitary or hypothalamic insufficiency (suspect if T_4 or T_3 are at the lower end of the reference range), use of TSH -suppressing medication, eg thyroxine, steroids.
- If $TSH < 0.1$, treat on an individual basis, eg with symptoms of hyperthyroidism, AF, unexplained weight loss, osteoporosis, goitre.
- Options are carbimazole or propylthiouracil—or radioiodine therapy.
- If no symptoms, recheck 6-monthly.



Fig 5.16 Facial appearance in hypothyroidism. Look for: pallor; coarse, brittle, diminished hair (scalp, axillary, and pubic); dull or blank expression lacking sparkle; coarse features; puffy lids. These signs are subtle: $\blacktriangleright$ have a low threshold for measuring TSH .

Reproduced from Cox and Roper, *Clinical Skills*, 2005, with permission from Oxford University Press.

Parathyroid hormone (PTH) is normally secreted in response to low ionized Ca^{2+} levels, by four parathyroid glands situated posterior to the thyroid (p601). The glands are controlled by $-ve$ feedback via Ca^{2+} levels. PTH acts by: $\bullet$ osteoclast activity releasing Ca^{2+} and PO_4^{3-} from bones $\bullet$ $\uparrow\text{Ca}^{2+}$ and $\downarrow\text{PO}_4^{3-}$ reabsorption in the kidney $\bullet$ active 1,25 dihydroxy-vitamin D_3 production is $\uparrow$. Overall effect is $\uparrow\text{Ca}^{2+}$ and $\downarrow\text{PO}_4^{3-}$.

Primary hyperparathyroidism *Causes:* $\sim 80\%$ solitary adenoma, $\sim 20\%$ hyperplasia of all glands, $<0.5\%$ parathyroid cancer. *Presentation:* Often 'asymptomatic' ($\blacktriangleright$ not in retrospect!), with $\uparrow\text{Ca}^{2+}$ on routine tests. Signs relate to: **1** $\uparrow\text{Ca}^{2+}$ (p676): weak, tired, depressed, thirsty, dehydrated-but-polyuric; also renal stones, abdominal pain, pancreatitis, and ulcers (duodenal:gastric $\approx 7:1$). **2** Bone resorption effects of PTH can cause pain, fractures, and osteopenia/osteoporosis. **3** $\uparrow\text{BP}$: $\blacktriangleright$ so check Ca^{2+} in *everyone* with hypertension. *Association:* MEN-1 (BOX 'Multiple endocrine neoplasia'). *Tests:* $\uparrow\text{Ca}^{2+}$ & $\uparrow\text{PTH}$ or inappropriately normal (other causes of this: thiazides, lithium, familial hypocalciuric hypercalcaemia, tertiary hyperparathyroidism). Also $\downarrow\text{PO}_4^{3-}$ (unless in renal failure), $\uparrow\text{ALP}$ from bone activity, 24h urinary $\uparrow\text{Ca}^{2+}$. *Imaging:* osteitis fibrosa cystica (due to severe resorption; rare) may show up as subperiosteal erosions, cysts, or brown tumours of phalanges $\pm$ acro-osteolysis (fig 5.17) $\pm$ 'pepper-pot' skull. *DEXA* (p683; for osteoporosis, p682). *Rx:* If mild: advise $\uparrow$ fluid intake to prevent stones; avoid thiazides + high Ca^{2+} & vit D intake; see 6-monthly. Excision of the adenoma or of all four hyperplastic glands prevents fractures and peptic ulcers. *Indications:* high serum or urinary Ca^{2+} , bone disease, osteoporosis, renal calculi, $\downarrow$ renal function, age ≤ 50 yrs. *Complications:* Hypoparathyroidism, recurrent laryngeal nerve damage ($\therefore$ hoarse), symptomatic $\text{Ca}^{2+}\downarrow$ (hungry bones syndrome; check Ca^{2+} daily for ≥ 14 d post-op). Pre-op US and MIBI scan may localize an adenoma; intra-operative PTH sampling is used to confirm removal. *Recurrence:* $\sim 8\%$ over 10 yrs.¹⁹ Cinacalcet (a 'calcimimetic') $\uparrow$ sensitivity of parathyroid cells to Ca^{2+} ($\therefore$ $\downarrow\text{PTH}$ secretion); monitor Ca^{2+} within 1 week of dose changes; SE: myalgia; $\downarrow$ testosterone.



Fig 5.17 Acro-osteolysis.

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Secondary hyperparathyroidism $\downarrow\text{Ca}^{2+}$, $\uparrow\text{PTH}$ (appropriately). *Causes:* $\downarrow$ vit D intake, chronic renal failure. *Rx:* Correct causes. Phosphate binders; vit D; cinacalcet if $\text{PTH} \geq 85 \text{ pmol/L}$ and parathyroidectomy tricky.

Tertiary hyperparathyroidism $\uparrow\text{Ca}^{2+}$, $\uparrow\text{PTH}$ (inappropriately). Occurs after prolonged secondary hyperparathyroidism, causing glands to act autonomously having undergone hyperplastic or adenomatous change. This causes $\uparrow\text{Ca}^{2+}$ from $\uparrow$ secretion of PTH unlimited by feedback control. Seen in chronic renal failure.

Malignant hyperparathyroidism Parathyroid-related protein (PTHrP) is produced by some squamous cell lung cancers, breast and renal cell carcinomas. This mimics PTH resulting in $\uparrow\text{Ca}^{2+}$ (PTH is $\downarrow$, as PTHrP is not detected in the assay).

Hypoparathyroidism

Primary hypoparathyroidism PTH secretion is $\downarrow$ due to gland failure. *Tests:* $\downarrow\text{Ca}^{2+}$, $\uparrow\text{PO}_4^{3-}$ or $\leftrightarrow$, $\leftrightarrow\text{ALP}$. *Signs:* Those of hypocalcaemia, p678 $\pm$ autoimmune comorbidities (BOX 'Autoimmune polyendocrine syndromes'). *Causes:* Autoimmune; congenital (Di George syn., OHCS p642). *Rx:* Ca^{2+} supplements + calcitriol (or synthetic PTH /12h sc: it prevents hypocalciuria).

Secondary hypoparathyroidism Radiation, surgery (thyroidectomy, parathyroidectomy), hypomagnesaemia (magnesium is required for PTH secretion).

Pseudohypoparathyroidism Failure of target cell response to PTH. *Signs:* Short metacarpals (esp. 4th and 5th, fig 5.18), round face, short stature, calcified basal ganglia (fig 5.19), $\uparrow\text{IQ}$. *Tests:* $\downarrow\text{Ca}^{2+}$, $\uparrow\text{PTH}$, $\leftrightarrow$ or $\uparrow\text{ALP}$. *Rx:* As for 1° hypoparathyroidism.

Pseudopseudohypoparathyroidism The morphological features of pseudohypoparathyroidism, but with normal biochemistry. The cause for both is genetic.

Multiple endocrine neoplasia (MEN types 1, 2a, and 2b)

In MEN syndromes there are functioning hormone-producing tumours in multiple organs (they are inherited as autosomal dominants).²⁰ They comprise: •MEN-1 and 2 •Neurofibromatosis (p514) •Von Hippel-Lindau and Peutz-Jeghers syndromes (p712 & p708) •Carney complex (spotty skin pigmentation, schwannomas, myxoma of skin, mucosa, or heart, especially atrial myxoma), and endocrine tumours: eg pituitary adenoma, adrenal hyperplasia, and testicular tumour.

MEN-1:

- Parathyroid hyperplasia/adenoma (~95%; most $\uparrow\text{Ca}^{2+}$).
- Pancreas endocrine tumours (70%)—gastrinoma (p716) or insulinoma (p215), or, rarely, somatostatinoma (DM + steatorrhoea + gallstones/cholangitis), VIPoma (p258), or glucagonomas ($\pm$ glucagon syndrome: migrating rash; glossitis; cheilitis, **fig 8.5** p327; anaemia; $\uparrow$ weight; $\uparrow$ plasma glucagon; $\uparrow$ glucose).
- Pituitary prolactinoma (~50%) or GH secreting tumour (acromegaly;²¹ p239); also, adrenal and carcinoid tumours are associated.

The MEN-1 gene is a tumour suppressor gene. Menin, its protein, alters transcription activation. Many are sporadic, presenting in the 3rd–5th decades.

MEN-2a:

- Thyroid: medullary thyroid carcinoma (seen in ~100%, p600).
- Adrenal: pheochromocytoma (~50%, usually benign and bilateral).
- Parathyroid hyperplasia (~80%, but less than 20% have $\uparrow\text{Ca}^{2+}$).

MEN-2b: Has similar features to MEN-2a plus mucosal neuromas and Marfanoid appearance (p706), but no hyperparathyroidism. Mucosal neuromas consist of 'bumps' on: lips, cheeks, tongue, glottis, eyelids, and visible corneal nerves.

The gene involved in MEN-2a and b is the ret proto-oncogene, a receptor tyrosine kinase. Tests for ret mutations are revolutionizing MEN-2 treatment by enabling a prophylactic thyroidectomy to be done before neoplasia occurs, usually before 3yrs of age. NB: ret mutations rarely contribute to sporadic parathyroid tumours.

Autoimmune polyendocrine syndromes

Autoimmune disorders cluster into two defined syndromes:

Type 1 Autosomal recessive, rare.

Cause: Mutations of **AIRE** (Auto Immune REgulator) gene on chromosome 21.

Features: •Addison's disease. •Chronic mucocutaneous candidiasis. •Hypoparathyroidism. Also associated with hypogonadism, pernicious anaemia, autoimmune primary hypothyroidism, chronic active hepatitis, vitiligo, alopecia.

Type 2 HLA D3 and D4 linked, common. **Cause:** Polygenic.

Features: •Addison's disease. •Type 1 diabetes mellitus (in 20%).

• Autoimmune thyroid disease—hypothyroidism or Graves' disease.

Also associated with primary hypogonadism, vitiligo, alopecia, pernicious anaemia, chronic atrophic gastritis, coeliac disease, dermatitis herpetiformis.



Fig 5.18 Pseudohypoparathyroidism: short 4th and 5th metacarpals.

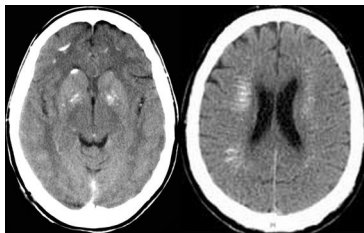


Fig 5.19 Cerebral calcification in pseudohypoparathyroidism: periventricular (left) and basal ganglia (right).

Courtesy of Professor Peter Scally.

Physiology The adrenal cortex produces steroids: **1 Glucocorticoids** (eg cortisol), which affect carbohydrate, lipid, and protein metabolism. **2 Mineralocorticoids**, which control sodium and potassium balance (eg aldosterone, p668). **3 Androgens**, sex hormones which have weak effect until peripheral conversion to testosterone and dihydrotestosterone. Corticotropin-releasing factor (CRF) from the hypothalamus stimulates ACTH secretion from the pituitary, which in turn stimulates cortisol and androgen production by the adrenal cortex. Cortisol is excreted as urinary free cortisol and various 17-oxogenic steroids.

Cushing's syndrome This is the clinical state produced by chronic glucocorticoid excess + loss of the normal feedback mechanisms of the hypothalamo-pituitary-adrenal axis and loss of circadian rhythm of cortisol secretion (normally highest on waking). The chief cause is oral steroids. Endogenous causes are rare: 80% are due to $\uparrow$ ACTH; of these a pituitary adenoma (Cushing's disease) is the commonest cause.

1 ACTH-dependent causes ($\uparrow$ ACTH)

- **Cushing's disease:** Bilateral adrenal hyperplasia from an ACTH-secreting pituitary adenoma (usually a microadenoma, p234). $\text{♀}:\text{♂} > 1:1$. Peak age: 30–50yrs. A low-dose dexamethasone test (Box) leads to no change in plasma cortisol, but 8mg may be enough to more than halve morning cortisol (as occurs in normals).
- **Ectopic ACTH production:** Especially small cell lung cancer and carcinoid tumours, p271. Specific features: pigmentation (due to $\uparrow$ ACTH), hypokalaemic metabolic alkalosis ($\uparrow$ cortisol leads to mineralocorticoid activity), weight loss, hyperglycaemia. Classical features of Cushing's are often absent. Dexamethasone even in high doses (8mg) fails to suppress cortisol production.
- **Rarely, ectopic CRF production:** Some thyroid (medullary) and prostate cancers.

2 ACTH-independent causes ($\downarrow$ ACTH due to $-ve$ feedback)

- **Iatrogenic:** Pharmacological doses of steroids (common).
- **Adrenal adenoma/cancer:** (May cause abdo pain $\pm$ virilization in ♀ , p230.) Because the tumour is autonomous, dexamethasone in any dose won't suppress cortisol.
- **Adrenal nodular hyperplasia:** (As for adrenal adenoma, no dexamethasone suppression.)
- **Rarely: Carney complex**, p223. **McCune-Albright syndrome**, see OHCS p650.

Symptoms $\uparrow$ Weight; mood change (depression, lethargy, irritability, psychosis); proximal weakness; gonadal dysfunction (irregular menses; hirsutism; erectile dysfunction); acne; recurrent Achilles tendon rupture; occasionally virilization if ♀ .

Signs Central obesity; plethoric, moon face; buffalo hump; supraclavicular fat distribution; skin & muscle atrophy; bruises; purple abdominal striae (fig 5.20); osteoporosis; $\uparrow$ BP; $\uparrow$ glucose; infection-prone; poor healing. Signs of the cause (eg abdo mass).

Tests Random plasma cortisols may mislead, as illness, time of day, and stress (eg venepuncture) influence results. Also, don't rely on imaging to localize the cause: non-functioning 'incidentalomas' occur in $\sim 5\%$ on adrenal CT and $\sim 10\%$ on pituitary MRI. MRI detects only $\sim 70\%$ of pituitary tumours causing Cushing's (many are too small).

Treatment Depends on the cause.

- **Iatrogenic:** Stop medications if possible.
- **Cushing's disease:** Selective removal of pituitary adenoma (trans-sphenoidally). Bilateral adrenalectomy if source unlocatable, or recurrence post-op (complication: *Nelson's syndrome*: $\uparrow$ skin pigmentation due to $\uparrow$ ACTH from an enlarging pituitary tumour, as adrenalectomy removes $-ve$ feedback; responds to pituitary radiation).
- **Adrenal adenoma or carcinoma:** Adrenalectomy: 'cures' adenomas but rarely cures cancer. Radiotherapy & adenylytic drugs (mitotane) follow if carcinoma.
- **Ectopic ACTH:** Surgery if tumour is located and hasn't spread. Metyrapone, ketoconazole, and fluconazole $\downarrow$ cortisol secretion pre-op or if awaiting effects of radiation. Intubation + mifepristone (competes with cortisol at receptors) + etomidate (blocks cortisol synthesis) may be needed, eg in severe ACTH-associated psychosis.

Prognosis Untreated Cushing's has $\uparrow$ vascular mortality.²² Treated, prognosis is good (but myopathy, obesity, menstrual irregularity, $\uparrow$ BP, osteoporosis, subtle mood changes and DM often remain—so follow up carefully, and manage individually).

Investigating suspected Cushing's syndrome

First, confirm the diagnosis (a raised plasma cortisol), then localize the source on the basis of laboratory testing. Use imaging studies to confirm the likely source.

1st-line tests *Overnight dexamethasone suppression test* is a good outpatient test. Dexamethasone 1mg PO at midnight; do serum cortisol at 8AM. Normally, cortisol suppresses to $<50\text{nmol/L}$; no suppression in Cushing's syndrome. False -ve rate: $<2\%$; false +ves: 2% normal, 13% obese, and 23% of inpatients. *NB:* false +ves (*pseudo-Cushing's*) are seen in depression, obesity, alcohol excess, and inducers of liver enzymes (trate of dexamethasone metabolism, eg phenytoin, phenobarbital, rifampicin, p689). *24h urinary free cortisol* (normal: $<280\text{nmol/24h}$) is an alternative.

2nd-line tests If 1st-line tests abnormal: *48h dexamethasone suppression test:* Give dexamethasone 0.5mg/6h PO for 2d. Measure cortisol at 0 and 48h (last test at 6h after last dose). Again, in Cushing's syndrome, there is a failure to suppress cortisol. *48h high-dose dexamethasone suppression test:* (2mg/6h.) May distinguish pituitary (suppression) from others causes (no/part suppression). *Midnight cortisol:* Admit (unless salivary cortisol used). Often inaccurate due to measurement issues. Normal circadian rhythm (cortisol *lowest* at midnight, *highest* early morning) is lost in Cushing's syndrome. Midnight blood, via a cannula during sleep, shows cortisol $\uparrow$ in Cushing's.

Localization tests (Where is the lesion?) If the 1st- and 2nd-line tests are +ve—*Plasma ACTH.* If ACTH is undetectable, an adrenal tumour is likely $\rightarrow$ CT/MRI adrenal glands. If no mass, proceed to *adrenal vein sampling.* If ACTH is detectable, distinguish a pituitary cause from ectopic ACTH production by high-dose suppression test or *corticotropin-releasing hormone (CRH) test:* 100mcg ovine or human CRH IV. Measure cortisol at 120min. Cortisol rises with pituitary disease but not with ectopic ACTH production.

If tests indicate that cortisol responds to manipulation, Cushing's disease is likely. Image the pituitary (MRI) and consider *bilateral inferior petrosal sinus blood sampling.*

If tests indicate that cortisol does not respond to manipulation, hunt for the source of ectopic ACTH—eg IV contrast CT of chest, abdomen, and pelvis $\pm$ MRI of neck, thorax, and abdomen, eg for small ACTH secreting carcinoid tumours.

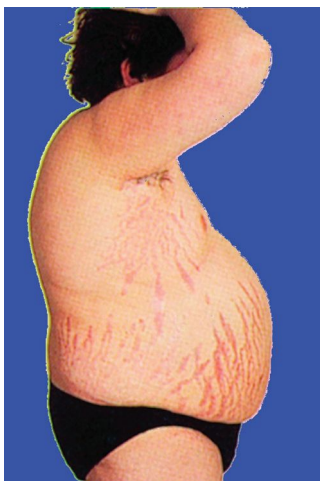


Fig 5.20 Hypercortisolism weakens skin; even normal stretching (or the pressure of obesity, as here) can make its elastin break—on healing we see these depressed purple scars (striae). Cortisone or rapid growth contributes to striae in other contexts: pregnancy, adolescence, weight lifting, sudden-onset obesity, or from strong steroid creams. Striae mature into silvery crescents looking like the underside of willow leaves. Unsightly immature striae may be improved by YAG lasers.

Primary adrenocortical insufficiency (Addison's disease) is rare (~0.8/100 000), but can be fatal. Destruction of the adrenal cortex leads to glucocorticoid (cortisol) and mineralocorticoid (aldosterone) deficiency (see [fig 5.21](#)). Signs are capricious: it is '*the unforgiving master of non-specificity and disguise*'.²³ You may diagnose a viral infection or anorexia nervosa in error (K^+ is $\downarrow$ in the latter but $\uparrow$ in Addison's).

Physiology:

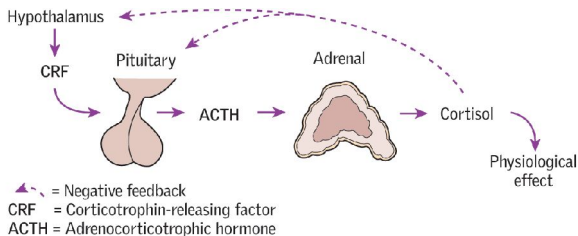


Fig 5.21 Pathways involved in adrenal function.

Causes: 80% are due to autoimmunity in the UK. **Other causes:** TB (commonest cause worldwide), adrenal metastases (eg from lung, breast, renal cancer), lymphoma, opportunistic infections in HIV (eg CMV, *Mycobacterium avium*, p400); adrenal haemorrhage ($\rightarrow$ Waterhouse-Friderichsen syndrome p714; antiphospholipid syndrome; SLE), congenital (late-onset congenital adrenal hyperplasia).

Secondary adrenal insufficiency The commonest cause is iatrogenic, due to long-term steroid therapy leading to suppression of the pituitary-adrenal axis. This only becomes apparent on withdrawal of the steroids. Other causes are rare and include hypothalamic-pituitary disease leading to $\downarrow$ ACTH production. Mineralocorticoid production remains intact, and there is no hyperpigmentation as $\downarrow$ ACTH.

Symptoms Often diagnosed late: lean, tanned, tired, tearful $\pm$ weakness, anorexia, dizzy, faints, flu-like myalgias/arthritis. **Mood:** depression, psychosis. **GI:** nausea/vomiting, abdominal pain, diarrhoea/constipation. **Think of Addison's in all with unexplained abdominal pain or vomiting.** Pigmented palmar creases & buccal mucosa ($\uparrow$ ACTH; cross-reacts with melanin receptors). Postural hypotension. Vitiligo. $\rightarrow$ **Signs of critical deterioration** (p836): Shock ($\downarrow$ BP, tachycardia), T° , coma.

Tests $\downarrow Na^+$ & $\uparrow K^+$ (due to $\downarrow$ mineralocorticoid), $\downarrow$ glucose (due to $\downarrow$ cortisol). Also: uraemia, $\uparrow Ca^{2+}$, eosinophilia, anaemia. Δ : **Short ACTH stimulation test** (Synacthen[®] test): Do plasma cortisol before and $\frac{1}{2}$ h after tetracosactide (Synacthen[®]) 250mcg IM. Addison's is excluded if 30min cortisol >550 nmol/L. Steroid drugs may interfere with assays: ask lab. NB: in pregnancy and contraceptive pill, cortisol levels may be reassuring but falsely $\uparrow$, due to $\uparrow$ cortisol-binding globulin. **Also:** $\bullet$ ACTH: In Addison's, 9AM ACTH is $\uparrow$ (>300 ng/L: inappropriately high). It is low in secondary causes $\bullet$ 21-Hydroxylase adrenal autoantibodies: +ve in autoimmune disease in $>80\%$ $\bullet$ Plasma renin & aldosterone: to assess mineralocorticoid status. **AXR/CXR:** Any past TB, eg upper zone fibrosis or adrenal calcification? If no autoantibodies, consider further tests (eg adrenal CT) for TB, histoplasma, or metastatic disease.

Treatment $\rightarrow$ See p836 for Addisonian crisis (shocked). Replace steroids: ~15–25mg hydrocortisone daily, in 2–3 doses, eg 10mg on waking, 5mg lunchtime. Avoid giving late (may cause insomnia). Mineralocorticoids to correct postural hypotension, $\downarrow Na^+$, $\uparrow K^+$: fludrocortisone PO from 50–200mcg daily. Adjust both on clinical grounds. If there is a poor response, suspect an associated autoimmune disease (check thyroid, do coeliac serology: p266).

Steroid use Advise wearing a bracelet declaring steroid use. Add 5–10mg hydrocortisone to daily intake before strenuous activity/exercise. Double steroids in febrile illness, injury, or stress. Give out syringes and in-date IM hydrocortisone, and show how to inject 100mg IM if vomiting prevents oral intake (seek medical help; admit for IV fluids if dehydrated).

Follow-up Yearly (BP, U&E); watch for autoimmune diseases (pernicious anaemia).⁴

Prognosis (treated) Adrenal crises and infections do cause excess deaths: mean age at death for men is ~65yrs (11yrs <estimated life expectancy; women lose ~3yrs).

⚠ Exogenous steroid use

Replacement steroids are vital in those taking long-term steroids when acutely unwell. Adrenal insufficiency may develop with deadly hypovolaemic shock, if additional steroid is not given. ▶▶ See p836.

Steroid use: Warn against abruptly stopping steroids. Emphasize that prescribing doctors/dentists/surgeons *must* know of steroid use: give *steroid card*.

Excerpt from the notes of Miss E.L.R., 92 days before her death from undiagnosed Addison's disease. From the Coroner's Court...

'Typical day—wakes up at 11.30, still feels tired, then will have some breakfast and usually fall asleep on the couch. The most energy req. activity in last 1 month is—cooking herself a pasta meal. Then, totally exhausted will sleep more in pm, then eat some dinner. Goes to bed at 11pm—latest. Not able to concentrate...Used to weigh 45kg. Now weighs 42kg.'²⁴

Placed on a page about Addison's disease, we might think there are sufficient clues to raise the suspicion of Addison's (even though her electrolytes were not particularly awry, and her pigmentation was barely perceptible). But change the context to our last busy clinic. We are a little distracted. The memory of Addison's is fading. Who among us will hear the alarm bell ring?

⁴ Autoimmune polyglandular syndromes types 1–4: **1** Monogenic syndrome (AIRE gene on chromosome 21); signs: candidiasis, hypoparathyroidism + Addison's. **2** (Schmidt syndrome.) Adrenal insufficiency + autoimmune thyroid disease ± DM ± pleuritis/pericarditis. **3** Autoimmune thyroid disease + other autoimmune conditions but *not* Addison's. **4** Autoimmune combinations not included in 1–3.

Primary hyperaldosteronism Excess production of aldosterone, independent of the renin-angiotensin system, causing ↑sodium and water retention, and ↓renin release. Consider if: hypertension, hypokalaemia, or alkalosis in someone not on diuretics. Sodium tends to be mildly raised or normal.

Symptoms: Often asymptomatic or signs of hypokalaemia (p674): weakness (even quadriparesis), cramps, paraesthesiae, polyuria, polydipsia. ↑BP but not always.

Causes: ~⅓ due to a solitary aldosterone-producing adenoma (linked to mutations in K^+ channels)⁵—*Conn's syndrome*. ~⅓ due to bilateral adrenocortical hyperplasia. Rare causes: adrenal carcinoma; or glucocorticoid-remediable aldosteronism (GRA)—the ACTH regulatory element of the 11β-hydroxylase gene fuses to the aldosterone synthase gene, ↑aldosterone production, & bringing it under the control of ACTH.

Tests: U&E, renin and aldosterone, and adrenal vein sampling. Do not rely on a low K^+ , as >20% are normokalaemic. For GRA (suspect if there is a family history of early hypertension), genetic testing is available. **Treatment:** •*Conn's*: laparoscopic adrenalectomy. Spironolactone (25–100mg/24h PO) for 4wks pre-op controls BP and K^+ . •*Hyperplasia*: treated medically: spironolactone or amiloride. •*GRA*: dexamethasone 1mg/24h PO for 4wks, normalizes biochemistry but not always BP. If BP is still ↑, use spironolactone as an alternative. •*Adrenal carcinoma*: surgery ± post-operative adrenolytic therapy with mitotane—prognosis is poor.

Secondary hyperaldosteronism Due to a high renin from ↓renal perfusion, eg in renal artery stenosis, accelerated hypertension, diuretics, CCF, or hepatic failure.

Bartter's syndrome This is a major cause of congenital (autosomal recessive) salt wasting—via a sodium and chloride leak in the loop of Henle via mutations in channels and transporters. Presents in childhood with failure to thrive, polyuria, and polydipsia. BP is normal. Sodium loss leads to volume depletion, causing ↑renin and aldosterone production, leading to hypokalaemia and metabolic alkalosis, urinary K^+ and Cl^- . **Treatment:** K^+ replacement, NSAIDs (to inhibit prostaglandins), and ACE-i.

Phaeochromocytoma

Rare catecholamine-producing tumours. They arise from sympathetic paraganglia cells (=phaeochromocytomas), which are collections of chromaffin cells. They are usually found within the adrenal medulla. Extra-adrenal tumours (paragangliomas) are rarer, and often found by the aortic bifurcation (the organs of Zuckerkandl). Phaeochromocytomas *roughly* follow the 10% rule: 10% are malignant, 10% are extra-adrenal, 10% are bilateral, and 10% are familial. Recent data suggest higher preponderance in patients with genetic mutations affecting several genes including SDH (succinyl dehydrogenase). Thus, family history is crucial and referral for genetic screening (particularly <50 years old). A dangerous but treatable cause of hypertension (in <0.1%). **Associations** ~90% are sporadic; 10% are part of hereditary cancer syndromes (p215), eg thyroid, MEN-2A and 2B, neurofibromatosis, von Hippel-Lindau syndrome (SDH mutations). **Classic triad** Episodic headache, sweating, and tachycardia (± ↑, ↓, or ↔BP, see BOX 'Features of phaeochromocytoma').

Tests •*Biochemical*: 24h urine for metanephrines/metadrenaline (better than catecholamines and vanillylmandelic acid²⁵), ↑WCC. •*Localization*: Abdominal CT/MRI, or meta-iodobenzylguanidine (chromaffin-seeking isotope) scan (can find extra-adrenal tumours, p738). **Treatment** *Surgery*: α-blockade pre-op: phenoxybenzamine (α-blocker) is used before β-blocker to avoid crisis from unopposed α-adrenergic stimulation, β-block too if heart disease or tachycardic. *Consult the anaesthetist*. *Post-op*: Do 24h urine metanephrine 2wks post-op, monitor BP (risk of ↓BP). **Emergency Rx**: p837. If malignant, chemotherapy or therapeutic radiolabelled MIBG may be used. **Follow-up**: Lifelong: malignant recurrence may present late, genetic screening.

5 Tumours from the zona glomerulosa, zona fasciculata, or zona reticularis associate with syndromes of ↑ mineralocorticoids, glucocorticoids, or androgens respectively, usually; remember 'GFR≈miner GA'.

Hypertension: a common context for hyperaldosteronism tests

Think of Conn's in these contexts:

- Hypertension associated with hypokalaemia.
- Refractory hypertension, eg despite ≥ 3 antihypertensive drugs.
- Hypertension occurring before 40yrs of age (especially in women).

The approach to investigation remains controversial, but the simplest is to look for a suppressed renin and $\uparrow$ aldosterone (may be normal if there is severe hypokalaemia). CT or MRI of the adrenals is done to localize the cause. This should be done after hyperaldosteronism is proven, due to the high number of adrenal incidentalomas. If imaging shows a unilateral adenoma, *adrenal vein sampling* may be done (venous blood is sampled from both adrenals). If one side reveals increased aldosterone:cortisol ratio compared with the other (>3 -fold difference), an adenoma is likely, and surgical excision is indicated. If no nodules or bilateral nodules are seen, think about adrenal hyperplasia or GRA.

►NB: renal artery stenosis is a more common cause of refractory $\uparrow$ BP and $\uparrow$ K⁺ (p315).

Features of pheochromocytoma (often episodic and often vague)

Try to diagnose before death: suspect if BP hard to control, accelerating, or episodic.

- **Heart:** $\uparrow$ Pulse; palpitations/VT; dyspnoea; faints; angina; MI/LVF; cardiomyopathy.⁶
- **CNS:** Headache; visual disorder; dizziness; tremor; numbness; fits; encephalopathy; Horner's syndrome (paraganglioma); subarachnoid/CNS haemorrhage.
- **Psychological:** Anxiety; panic; hyperactivity; confusion; episodic psychosis.
- **Gut:** D&V; abdominal pain over tumour site; mass; mesenteric vasoconstriction.
- **Others:** Sweats/flushes; heat intolerance; pallor; $\uparrow$ T[°]; backache; haemoptysis.

Symptoms may be precipitated by straining, exercise, stress, abdominal pressure, surgery, or by agents such as β -blockers, IV contrast agents, or the tricyclics. The site of the tumour may determine precipitants, eg if pelvic, precipitants include sexual intercourse, parturition, defecation, and micturition. Adrenergic crises may last minutes to days. Suddenly patients feel 'as if about to die'—and then get better, or go on to develop a stroke or cardiogenic shock. On examination, there may be no signs, or hypertension $\pm$ signs of heart failure/cardiomyopathy ($\pm$ paradoxical shock, similar to Takotsubo's⁶), episodic thyroid swelling, glycosuria during attacks, or terminal haematuria from a bladder pheochromocytoma.

⁶ *Takotsubo cardiomyopathy* (=stress- or catecholamine-induced cardiomyopathy/broken heart syndrome) may cause sudden chest pain mimicking MI, with $\uparrow$ ST segments, and its signature apical ballooning on echo (also $\uparrow$ ejection fraction) occurring during catecholamine surges. It is a cause of MI in the presence of normal arteries. The stress may be medical (SAH, p478) or psychological.

Hirsutism is common (10% of women) and usually benign. It implies male pattern hair growth in women. Causes are familial, idiopathic, or are due to androgen secretion by the *ovary* (eg polycystic ovarian syndrome, ovarian cancer, *OHCS* p281), the *adrenal gland* (eg late-onset congenital adrenal hyperplasia, *OHCS* p251, Cushing's syndrome, adrenal cancer), or *drugs* (eg steroids). **Polycystic ovarian syndrome (PCOS)** causes secondary oligo- or amenorrhoea, infertility, obesity, acne, and hirsutism (*OHCS* p252). Ultrasound: bilateral polycystic ovaries. Blood tests: ↑testosterone (if $\geq 6\text{nmol/L}$, look for an androgen-producing adrenal or ovarian tumour), ↓sex-hormone binding globulin, tLH:FSH ratio (not consistent), TSH, lipids. Address any feelings of lack of conformity to society's perceived norms of feminine beauty.

Management: Healthy eating, optimize weight, shaving; laser photoepilation; wax; creams, eg eflornithine, or electrolysis (expensive/time-consuming, but effective); bleach (1:10 hydrogen peroxide).

- Oestrogens: combined contraceptive pill (*OHCS* p302)—Yasmin® is one choice as its progestogen, drospirenone, is an antimineralocorticoid. Alternatively, co-cyprindiol provided there are no contraindications, such as uncontrolled hypertension and current breast cancer. Stop co-cyprindiol 3–4 months after hirsutism has completely resolved because of increased VTE risk. If COCs are contraindicated or have not worked (after 6/12), refer the woman to secondary care for specialist treatment.
- Metformin (helps with insulin resistance) and spironolactone are sometimes tried.
- Clomifene is used for infertility (a fertility expert should prescribe).

Virilism Onset of amenorrhoea, clitoromegaly, deep voice, temporal hair recession + hirsutism. Look for an androgen-secreting adrenal or ovarian tumour.

Gynaecomastia (ie abnormal amount of breast tissue in men; may occur in normal puberty.) Oestrogen/androgen ratio↑ (vs galactorrhoea in which prolactin is↑).

Causes: Hypogonadism (see BOX 'Male hypogonadism'), liver cirrhosis (toestrogens), hyperthyroidism, tumours (oestrogen-producing, eg testicular, adrenal; HCG-producing, eg testicular, bronchial); drugs: oestrogens, spironolactone, digoxin, testosterone, marijuana; if stopping is impossible, consider testosterone if hypogonadism ± anti-oestrogen (tamoxifen).

Impotence (= erectile dysfunction) Erections result from neuronal release of nitric oxide (NO) which, via cGMP and Ca^{2+} , hyperpolarizes and thus relaxes vascular and trabecular smooth muscle cells, allowing engorgement. Common after 50yrs, and often multifactorial. A psychological facet is common (esp. if erectile dysfunction occurs only in some situations, if onset coincides with stress, and if early morning erections still occur: these also persist in early organic disease).

Organic causes: The big three: smoking, alcohol, and diabetes (reduce NO + autonomic neuropathy). Also *endocrine*: hypogonadism, hyperthyroidism, ↑prolactin; *neurological*: cord lesions, MS, autonomic neuropathy; *pelvic surgery*, eg bladder-neck, prostate; *radiotherapy*; *atheroma*; *renal or hepatic failure*; *prostatic hyperplasia*; *penile anomalies*, eg post-priapism, or Peyronie's (p708); *drugs*: digoxin, β -blockers, diuretics, antipsychotics, antidepressants, oestrogens, finasteride, narcotics.

Workup: After a full sexual and psychological history do: U&E, LFT, glucose, TFT, LH, FSH, lipids, testosterone, prolactin ± Doppler. Is penile arterial pressure enough for inflow? Is penile sensation OK (if not, ?CNS problem)? Is the veno-occlusive mechanism OK?

Rx: •Treat causes. •Counselling. •Oral phosphodiesterase (PDE5) inhibitors t cGMP. Erection isn't automatic (depends on erotic stimuli). Sildenafil 25–100mg $\frac{1}{2}$ –1h pre-sex (food and alcohol upset absorption). **SE:** Headache (16%); flushing (10%); dyspepsia (7%); stuffy nose (4%); transient blue-green tingeing of vision (inhibition of retinal PDE6). **CI:** See BOX 'Contraindications and cautions to PDE5 Inhibitors'. Tadalafil (long $t\frac{1}{2}$) 10–20mg $\frac{1}{2}$ –36h pre-sex. Don't use >once daily. Vardenafil (5–20mg). •Vacuum aids (ideal for penile rehabilitation after radical prostatectomy), intracavernosal injections, transurethral pellets, and prostheses (inflatable or malleable; partners may receive unnatural sensations). •Corpus cavernosum tissue engineering (eg on acellular collagen scaffolds) is in its infancy.

Contraindications and cautions to PDE5 inhibitors

►► Contraindications

- Concurrent use of nitrates.
- BP high or systolic <90mmHg/arrhythmia.
- Degenerative retinal disorders, eg retinitis pigmentosa.
- Unstable angina/stroke <6 months ago.
- Myocardial infarction <90 days ago.

Cautions ► Angina (especially if during intercourse).

- Bleeding; peptic ulcer (sildenafil).
- Marked hepatic or renal impairment.
- Peyronie's disease or cavernosal fibrosis.
- Risk of priapism (sickle-cell anaemia, myeloma, leukaemia).
- Concurrent complex antihypertensive regimens.
- Dyspnoea on minimal effort (sexual activity may be unsupportable).

Use in coronary artery disease has been a question, but is probably OK.²⁶

Interactions: Nitrates (contraindication); cytochrome p450 (CYP3A) inducers: macrolides, protease inhibitors, theophyllines, azole antifungals, rifampicin, phenytoin, carbamazepine, phenobarbital, grapefruit juice (bioavailability). Caution if α -blocker use; avoid vardenafil with type 1A (eg quinidine; procainamide) and type 3 anti-arrhythmics (sotalol; amiodarone)—as well as nitrates as above-mentioned.

Male hypogonadism

Hypogonadism is failure of testes to produce testosterone, sperm, or both. Features: small testes, ↓libido, erectile dysfunction, loss of pubic hair, ↓muscle bulk, ↑fat, gynaecomastia, osteoporosis, ↓mood. If prepubertal: ↓virilization; incomplete puberty; eunuchoid body; reduced secondary sex characteristics. Causes include:

Primary hypogonadism is due to testicular failure, eg from •local trauma, torsion, chemotherapy/irradiation •post-orchitis, eg mumps, HIV, brucellosis, leprosy •renal failure, liver cirrhosis, or alcohol excess (toxic to Leydig cells) •chromosomal abnormalities, eg Klinefelter's syndrome (47XXY)—delayed sexual development, small testes, and gynaecomastia. Anorchia is rare.

Secondary hypogonadism ↓Gonadotropins (LH & FSH), eg from •hypopituitarism •prolactinoma •Kallman's syndrome—isolated gonadotropin-releasing hormone deficiency, often with anosmia and colour blindness •systemic illness (eg COPD; HIV; DM) •Laurence-Moon-Biedl and Prader-Willi syndromes (OHCS p648 & p652) •Age.

R_x: (p232) If total testosterone ≤8nmol/L, on 2 mornings (or <15 if LH too) and ↓muscle bulk, testosterone may help, eg 1% dermal gel (Testogel®). Heart, bladder, and sexual function may perk up in age-related hypogonadism. Beware medicalizing ageing!

CI ↑Ca²⁺; nephrosis; polycythaemia; prostate, breast[♂] or liver ca. Monitor PSA.

Hypopituitarism entails ↓secretion of anterior pituitary hormones (figs 5.3, 5.22). They are affected in this order: growth hormone (GH), gonadotropins: follicle-stimulating hormone (FSH) and luteinizing hormone (LH), thyroid-stimulating hormone (TSH), and adrenocorticotrophic hormone (ACTH), prolactin (PRL). Panhypopituitarism is deficiency of all anterior hormones, usually caused by irradiation, surgery, or pituitary tumour.

Causes are at three levels. **1 Hypothalamus:** Kallman's syndrome (p231), tumour, inflammation, infection (meningitis, TB), ischaemia. **2 Pituitary stalk:** Trauma, surgery, mass lesion (craniopharyngioma, p234), meningioma, carotid artery aneurysm. **3 Pituitary:** Tumour, irradiation, inflammation, autoimmunity,⁷ infiltration (haemochromatosis, amyloid, metastases), ischaemia (pituitary apoplexy, p234; DIC⁸; Sheehan's syndrome⁹).

Features are due to:

1 Hormone lack: •**GH:** central obesity, atherosclerosis, dry wrinkly skin, ↓strength, ↓balance, ↓well-being, ↓exercise ability, ↓cardiac output, osteoporosis, ↓glucose. •**Gonadotropin (FSH; LH):** ♀ oligomenorrhoea or amenorrhoea, ↓fertility, ↓libido, osteoporosis, breast atrophy, dyspareunia. ♂ Erectile dysfunction, ↓libido, ↓muscle bulk, hypogonadism (↓hair, all over; small testes; ↓ejaculate volume; ↓spermatogenesis). •**Thyroid:** as for hypothyroidism (p220). •**Corticotropin:** as for adrenal insufficiency (p226). NB: no ↑skin pigmentation as ↓ACTH. •**Prolactin:** rare: absent lactation.

2 Causes: Eg pituitary tumour (p234), causing mass effect, or hormone secretion with ↓secretion of other hormones—eg prolactinoma, acromegaly, rarely Cushing's.

Tests (The triple stimulation test is now rarely done.)

- **Basal tests:** LH and FSH (↓ or ↔), testosterone or oestradiol (↓); TSH (↓ or ↔), T₄ (↓); prolactin (may be ↑, from loss of hypothalamic dopamine that normally inhibits its release), insulin-like growth factor-1 (IGF-1; ↓—used as a measure of GH axis, p238), cortisol (↓). Also do U&E (↓Na⁺ from dilution), ↓Hb (normochromic, normocytic).
- **Dynamic tests:** **1** Short Synacthen® test: (p226) to assess the adrenal axis. **2** Insulin tolerance test (ITT): done in specialist centres to assess the adrenal and GH axes. CI: epilepsy, heart disease, adrenal failure. Consult lab first. It involves IV insulin to induce hypoglycaemia, causing stress to ↑cortisol and GH secretion. It is done in the morning (water only taken from 22:00h the night before). Have 50% glucose and hydrocortisone to hand and IV access. Glucose must fall below 2.2mmol/L and the patient should become symptomatic when cortisol and GH are taken. Normal: GH >20mu/L, and peak cortisol >550nmol/L.
- 3** Arginine + growth hormone-releasing hormone test.
- 4** Glucagon stimulation test is alternative when ITT is contraindicated.
- **Investigate cause:** MRI scan to look for a hypothalamic or pituitary lesion.

Treatment Refer to an endocrinologist for assessment of pituitary function and to oversee hormone replacement and treatment of underlying cause.

- **Hydrocortisone** for 2° adrenal failure (p226) ▶ before other hormones are given.
- Thyroxine if hypothyroid (p220, but TSH is useless for monitoring).
- Hypogonadism (for symptoms and to prevent osteoporosis). ♂: options include *testosterone enanthate* 250mg IM every 3 weeks, daily topical gels or buccal mucoadhesive tablets. Patches (eg Testogel®) are also used. ♀: (premenopausal). *Oestrogen*: transdermal oestradiol patches, or contraceptive pill (exceeds replacement needs) ± testosterone or dehydroepiandrosterone (DHEA, in hypoandrogenic women; a small amount may improve well-being and sexual function, and help bone mineral density and lean body mass).
- **Gonadotropin** therapy is needed to induce fertility in both men and women.
- Growth hormone (GH). Somatotrophin mimics human GH. It addresses problems of ↑fat mass, ↓bone mass, ↓lean body mass (muscle bulk), ↓exercise capacity, and problems with heat intolerance.

⁷ Autoimmune hypophysitis (=inflamed pituitary) mimics pituitary adenoma. It may be triggered by pregnancy or immunotherapy blocking CTLA-4. No pituitary auto-antigen is yet used diagnostically.

⁸ Snake bite is a common cause in India (eg when associated with acute kidney injury).

⁹ Sheehan's syndrome is pituitary necrosis after postpartum haemorrhage.



Fig 5.22 Neuroendocrinology: emotions $\rightleftharpoons$ thoughts $\rightleftharpoons$ actions. As Michelangelo foretold (in his *Creation of Adam*) 'all gods and demons that have ever existed are within us as possibilities, desires, and ways of escape'. Within the dark red vault of our skull we see human and god-like forms reaching out, as thoughts escape into actions—with legs extending into our brainstem (**B**) and a fist is pushing from our hypothalamus into the pituitary stalk (**P**). Above the pituitary we have thoughts, ideas, impulses, and neurotransmitters. Below we have hormones. Between is the realm of neuroendocrinology—the neurosecretory cells which turn emotions into the releasing factors for the pituitary hormones (**fig 5.4**).

Image courtesy of Gary Bevans; quote from Frank Lynn Meshberger. *Michelangelo, Renaissance Man of the Brain, Too?*



Pituitary tumours (almost always benign adenomas) account for 10% of intracranial tumours (see [figs 5.23, 5.24](#)). They may be divided by size: a microadenoma is a tumour <1cm across, and a macroadenoma is >1cm. There are three histological types ([table 5.4](#)):

- 1 **Chromophobe** 70%. Many are non-secretory,¹⁰ some cause hypopituitarism. Half produce prolactin (PRL); a few produce ACTH or GH. Local pressure effect in 30%.
- 2 **Acidophil** 15%. Secrete GH or PRL. Local pressure effect in 10%.
- 3 **Basophil** 15%. Secrete ACTH. Local pressure effect rare.

Symptoms are caused by pressure, hormones (eg galactorrhoea), or hypopituitarism (p232). FSH-secreting tumours can cause macro-orchidism in men, but are rare.

Features of local pressure Headache, visual field defects (bilateral temporal hemianopia, due to compression of the optic chiasm), palsy of cranial nerves III, IV, VI (pressure or invasion of the cavernous sinus; [fig 5.25](#)). Also, diabetes insipidus (DI) (p240; more likely from hypothalamic disease); disturbance of hypothalamic centres of T°, sleep, and appetite; erosion through floor of sella leading to CSF rhinorrhoea.

Tests MRI defines intra- and supra-sellar extension; accurate assessment of visual fields; screening tests: PRL, IGF-1 (p238), ACTH, cortisol, TFTs, LH/FSH, testosterone in ♂, short Synacthen® test. Glucose tolerance test if acromegaly suspected (p238). If Cushing's suspected, see p225. Water deprivation test if DI is suspected (p240).

Treatment Start hormone replacement as needed (p232). Ensure steroids are given before levothyroxine, as thyroxine may precipitate an adrenal crisis. For Cushing's disease see p225, prolactinoma p236, acromegaly p238.

- **Surgery:** ([fig 5.26](#)) Most pituitary surgery is trans-sphenoidal, but if there is supra-sellar extension, a trans-frontal approach may be used. For prolactinoma, 1st-line treatment is medical with a dopamine agonist, p236. *Pre-op:* ensure hydrocortisone 100mg IV/IM. Subsequent cortisol replacement and reassessment varies with local protocols: get advice. *Post-op:* retest pituitary function (p232) to assess replacement needs. Repeating dynamic tests for adrenal function ≥6 weeks post-op.
- **Radiotherapy:** (Eg stereotactic.) Good for residual or recurrent adenomas (good rates of tumour control and normalization of excess hormone secretion).²⁷

Post-op Recurrence may occur late after surgery, so life-long follow-up is required. Fertility should be discussed: this may be reduced post-op due to ↓gonadotropins.

Pituitary apoplexy Rapid pituitary enlargement from a bleed into a tumour may cause mass effects, cardiovascular collapse due to acute hypopituitarism, and death. Suspect if acute onset of headache, meningism, ↓GCS, ophthalmoplegia/visual field defect, especially if there is a known tumour (may present like subarachnoid haemorrhage). **R:** Urgent steroids (hydrocortisone 100mg IV) and meticulous fluid balance ± cabergoline (dopamine agonist, if prolactinoma) ± surgery; find the cause, eg a predisposition to thrombosis, from antiphospholipid syndrome.

Craniopharyngioma Not strictly a pituitary tumour: it originates from Rathke's pouch so is situated between the pituitary and 3rd ventricle floor. They are rare, but are the commonest childhood intracranial tumour. Over 50% present in childhood with growth failure; adults may present with amenorrhoea, ↓libido, hypothalamic symptoms (eg DI, hyperphagia, sleep disturbance) or tumour mass effect. **Tests:** CT/MRI (calcification in 50%, may also be seen on skull x-ray). **Treatment:** Surgery ± post-op radiation; test pituitary function post-op.

Table 5.4 Frequency of hormones secreted by pituitary adenomas based on immunohistochemistry

Hormone	%	Hormone	%
PRL only (→prolactinoma)	35%	ACTH (→Cushing's disease)	7%
GH only (→acromegaly)	20%	LH/FSH/TSH	≥1%*
PRL and GH	7%	No obvious hormone	30%

*Sensitive methods of TSH measurement have improved recognition of TSH-secreting tumours. These are now more frequently found at microadenoma stage, medially located, and *without* associated hormone hypersecretion. In these tumours, somatostatin analogues (p238) are very helpful. See also Socin *et al.* *Eur J Endocrinol.* 2003;148:433-42.

10 If <1cm, usually 'incidentaloma'; most non-functioning macroadenomas are revealed by mass effect and/or hypopituitarism. Here, recurrence after surgery is common, so follow carefully with MRIs.

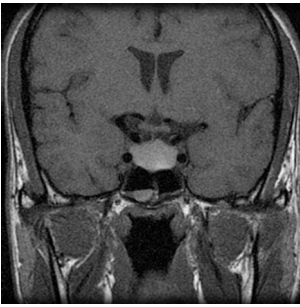


Fig 5.23 Sagittal T1-weighted MRI of the brain (no gadolinium contrast) showing a lesion in the pituitary fossa, most likely a haemorrhagic pituitary adenoma. Differential diagnosis includes a Rathke's cleft cyst.

Courtesy of Norwich Radiology Dept.

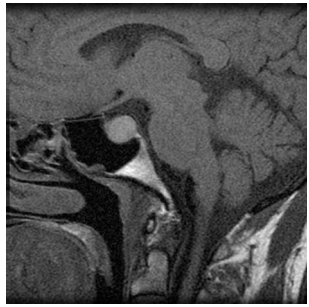


Fig 5.24 Coronal T1-weighted MRI of the brain (no gadolinium contrast) showing a lesion in the pituitary fossa (see **fig 5.23**).

Courtesy of Norwich Radiology Dept.

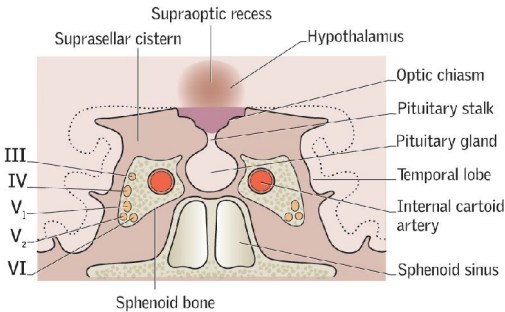


Fig 5.25 The pituitary gland's relationships to cranial nerves III, IV, V, and VI.

Reproduced from Turner and Wass, *Oxford Handbook of Endocrinology and Diabetes*, 2009, with permission from Oxford University Press.

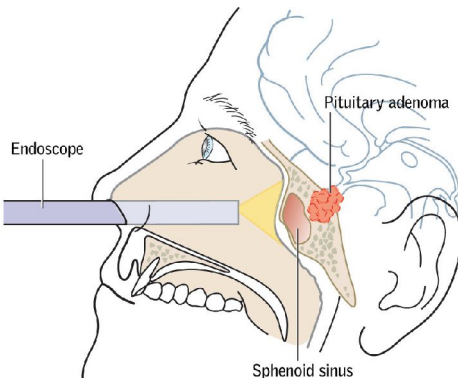


Fig 5.26 Endoscopic surgery is now possible for pituitary surgery.

This is the commonest hormonal disturbance of the pituitary. It presents earlier in women (menstrual disturbance) but later in men (eg with erectile dysfunction and/or mass effects). Prolactin stimulates lactation.^{11,12} Raised levels lead to hypogonadism, infertility, and osteoporosis, by inhibiting secretion of gonadotropin-releasing hormone (hence ↓LH/FSH and ↓testosterone or oestrogen).

Causes of raised plasma prolactin (PRL; >390mu/L) PRL is secreted from the anterior pituitary and release is inhibited by dopamine produced in the hypothalamus. Hyperprolactinaemia may result from **1** Excess production from the pituitary, eg prolactinoma. **2** Disinhibition, by compression of the pituitary stalk, reducing local dopamine levels. **3** Use of a dopamine antagonist. A PRL of 1000–5000mu/L may result from any, but >5000 is likely to be due to a prolactinoma, with macroadenomas (>10mm) having the highest levels, eg 10 000–100 000.

- **Physiological:** Pregnancy; breastfeeding; stress. Acute rises occur post-orgasm.¹¹
- **Drugs (most common cause):** Metoclopramide; haloperidol; methyldopa; oestrogens; ecstasy/MDMA;¹² antipsychotics (a reason for 'non-compliance': sustained hyperprolactinaemia may cause ↓libido, anorgasmia, and erectile dysfunction).
- **Diseases:** *Prolactinoma:* micro- or macroadenoma; *Stalk damage:* pituitary adenomas, surgery, trauma; *Hypothalamic disease:* craniopharyngioma, other tumours; *Other:* hypothyroidism (due to ↑TRH), chronic renal failure (↓excretion).

Symptoms ♀: Amenorrhoea or oligomenorrhoea; infertility; galactorrhoea (fig 5.27). Also: ↓libido, tweight, dry vagina. ♂: Erectile dysfunction, ↓facial hair, galactorrhoea. May present late with osteoporosis or local pressure effects from the tumour (p234).

Tests Basal PRL: non-stressful venepuncture between 09.00 and 16.00h. Do a pregnancy test, TFT, U&E. MRI pituitary if other causes are ruled out.

Management Refer to a specialist endocrinology clinic. Dopamine agonists (bromocriptine or cabergoline) are 1st line.

Microprolactinomas: A tumour <10mm on MRI (~25% of us have asymptomatic microprolactinomas). Bromocriptine, a dopamine agonist, ↓PRL secretion, restores menstrual cycles and ↓tumour size. Dose is titrated up: 1.25mg PO; increase weekly by 1.25–2.5mg/d until ~2.5mg/12h. SE: nausea, depression, postural hypotension (minimize by giving at night). If pregnancy is planned, use barrier contraception until 2 periods have occurred. If subsequent pregnancy occurs, stop bromocriptine after the 1st missed period. An alternative dopamine agonist is cabergoline: more effective and fewer SE, but there are fewer data on safety in pregnancy. NB: ergot alkaloids (bromocriptine and cabergoline) can cause fibrosis (eg echocardiograms are needed). Trans-sphenoidal surgery may be considered if intolerant of dopamine agonists. It has a high success rate, but there are risks of permanent hormone deficiency and prolactinoma recurrence, and so it is usually reserved as a 2nd-line treatment.

Macroprolactinomas: A tumour >10mm diameter on MRI. As they are near the optic chiasm, there may be ↓acuity, diplopia, ophthalmoplegia, visual-field loss, and optic atrophy. Treat initially with a dopamine agonist (bromocriptine if fertility is the goal). Surgery is rarely needed, but consider if visual symptoms or pressure effects which fail to respond to medical treatment. Bromocriptine, and in some cases radiation therapy, may be required post-op as complete surgical resection is uncommon. If pregnant, monitor closely ideally in a combined endocrine/antenatal clinic as there is risk of expansion.

Follow-up Monitor PRL. If headache or visual loss, check fields (? do MRI). Medication can be decreased after 2yrs, but recurrence of hyperprolactinaemia and expansion of the tumour may occur, and so these patients should be monitored carefully.

11 The prolactin increase (♂ and ♀) after coitus is ~400% greater than after masturbation; post-orgasmic prolactin is part of a feedback loop decreasing arousal by inhibiting central dopaminergic processes. The size of post-orgasmic prolactin increase is a neurohormonal index of sexual satisfaction.

12 MDMA also ↑oxytocin; prolactin + oxytocin are thought to mediate post-orgasmic well-being.



Fig 5.27 Galactorrhoea can be prolific enough to create medium-sized galaxies (bottom right). In the *Birth of the Milky Way* Hera is depicted by Rubens in her chariot, being drawn through the night sky by ominous black peacocks. Between journeys, she enjoyed discussing difficult endocrinological topics with her husband Zeus (who was also her brother), such as whether women or men find sexual intercourse more enjoyable. Hera inclined to the latter—and it is on this flimsy evidence, and her gorgeous galactorrhoea, that we diagnose her hyperprolactinaemia (which is known to decrease desire, lubrication, orgasm, and satisfaction). In the end, this issue was settled, in favour of Zeus's view, by Tiresias, who had unique insight into this intriguing question: every time this soothsayer saw two snakes entwined, (s)he changed sex, so coming to know a thing or two about gender and pleasure. This is a primordial example of an 'N of 1' trial, where the subject is his or her own control. Generalizability can be a problem with this methodology.

© Fine Art Images/ Age Fotostock.



This is due to tsecretion of GH (growth hormone) from a pituitary tumour (99%) or hyperplasia, eg via ectopic GH-releasing hormone from a carcinoid tumour. $\frac{GH}{IGF-1} \approx 1:1$. Incidence: UK 3/million/yr. ~5% are associated with MEN-1 (p223). GH stimulates bone and soft tissue growth through tsecretion of insulin-like growth factor-1 (IGF-1).

Symptoms Acroparaesthesia (*akron*=extremities); amenorrhoea; libido; headache; tsweating; snoring; arthralgia; backache; **fig 5.28**: 'My rings don't fit, nor my old shoes, and now I've got a wonky bite (malocclusion) and curly hair. I put on lots of weight, all muscle and looked good for a while; now I look so haggard'.

Signs (BOX 'Signs of acromegaly'). Often predate diagnosis by >4yrs. If acromegaly occurs before bony epiphyses fuse (rare), gigantism occurs.

Complications (May present with ccf or ketoacidosis.)

- Impaired glucose tolerance (~40%), DM (~15%).
- Vascular: tBP, left ventricular hypertrophy ($\pm$ dilatation/ccf), cardiomyopathy, arrhythmias. There is trisk of ischaemic heart disease and stroke (?due to tBP $\pm$ insulin resistance and GH-induced increase in fibrinogen and decrease in protein S).
- Neoplasia: tcolon cancer risk; colonoscopy may be needed.²¹

Acromegaly in pregnancy (Subfertility is common.) Pregnancy may be normal; signs and chemistry may remit. Monitor glucose.

Tests tGlucose, tCa²⁺, and tPO₄³⁻. **GH**: Don't rely on random GH as secretion is pulsatile and during peaks acromegalic and normal levels overlap. GH also t in: stress, sleep, puberty, and pregnancy. Normally GH secretion is inhibited by high glucose, and GH hardly detectable. In acromegaly GH release fails to suppress.

- If basal serum GH is >0.4mcg/L (1.2mIU/L) and/or if tIGF-I (p232), an oral glucose tolerance test (OGTT) is needed. If the lowest GH value during OGTT is above 1mcg/L (3mIU/L), acromegaly is confirmed. With general use of very sensitive assays, it has been said that this cut-off be decreased to 0.3mcg/L (0.9mIU/L).²⁸

Method: Collect samples for GH glucose at: 0, 30, 60, 90, 120, 150min. Possible false +ves: puberty, pregnancy, hepatic and renal disease, anorexia nervosa, and DM.

- MRI scan of pituitary fossa. • Look for hypopituitarism (p232).
- Visual fields and acuity. • ECG, echo. Old photos if possible.

Treatment Aim to correct (or prevent) tumour compression by excising the lesion, and to reduce GH and IGF-I levels to at least a 'safe' GH level of <2mcg/L (<6mIU/L). A 3-part strategy: **1** Transphenoidal surgery is often 1st line. **2** If surgery fails to correct GH/IGF-I hypersecretion, try somatostatin analogues (SSA) and/or radiotherapy, SSA being generally preferred. Example: octreotide (Sandostatin LAR®, given monthly IM), or lanreotide (Somatuline LA®). SE: pain at the injection site; gastrointestinal: abdominal cramps, flatulence, loose stools, tgallstones; impaired glucose tolerance. **3** The GH antagonist pegvisomant (recombinant GH analogue) is used if resistant or intolerant to SSA. It suppresses IGF-1 to normal in 90%, but GH levels may rise; rarely tumour size increases, so monitor closely. **Radiotherapy**: If unsuited to surgery or as adjuvant; may take years to work. **Follow-up**: Yearly GH, IGF-1 $\pm$ OGTT; visual fields; vascular assessment. BMI; photos (**fig 5.29**).

Prognosis May return to normal (any excess mortality is mostly vascular). 16% get diabetes with SSAs vs ~13% after surgery.



Fig 5.28 Acromegaly. Courtesy of Omar Rio.

My life with acromegaly. (<http://odelrio.blogspot.com>)

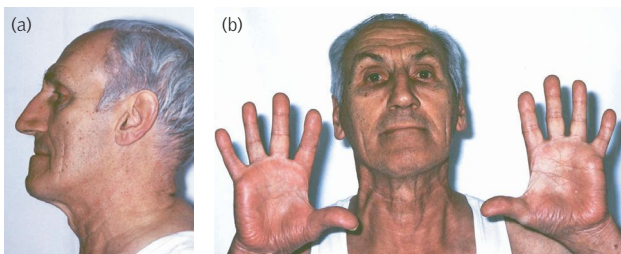


Fig 5.29 (a) and (b) Coarsening of the face and ↑ growth of hands and mandible (prognathism).

Signs of acromegaly

- ↑Growth of hands (**fig 5.29b**; may be spade-like), jaw (**fig 5.29a**) and feet (sole may encroach on the dorsum).
- Coarsening face; wide nose.
- Big supraorbital ridges.
- Macroglossia (big tongue).
- Widely spaced teeth.
- Puffy lips, eyelids, and skin (oily and large-pored); also skin tags.
- Scalp folds (*cutis verticis gyrata*; due to expanding but tethered skin).
- Skin darkening (**fig 5.29**).
- Acanthosis nigricans (**fig 12.28**, p563).
- Laryngeal dyspnoea (fixed cords).
- Obstructive sleep apnoea.
- Goitre (↑thyroid vascularity).
- Proximal weakness + arthropathy.
- Carpal tunnel signs in 50%, p503.
- Signs from any pituitary mass: hypopituitarism ± local mass effect (p232; ↓vision; hemianopia); fits.

Dysmorphia, personal identity, and acromegaly

We might have devoted this box to a grotesque homunculus depicting the signs of acromegaly: all disconnected lips, hands, feet, brows, and noses. But our integrative ethics disallow this, and ask us instead to see if acromegaly can reveal something universal about our patients and ourselves. What is it like to feel in the grip of some 'alien puberty' or 'empty pregnancy'? These analogies are physiological as well as metaphorical.^{14,15} The changes of acromegaly are not so insidious that the patient thinks all is fine: there is often partial knowledge and a few dark thoughts on looking into the mirror. Even when we lay our lives end-to-end for inspection (**fig 5.28**), changes are subtle. It can take the observations of others to force us to come face-to-face with the truth of our new unfolding self. In one patient the comment was 'So are you pregnant again?' 'Why do you ask?' 'Because your nose is as big as it was when you were last pregnant'. So here we have the well-known 'physiological acromegaly of pregnancy'¹⁴ predating the pathological, as the carnival of personal identity moves from helter-skelter to roller-coaster.

14 GH variants made by the placenta rise exponentially until 37wks' gestation; pituitary GH gradually drops to near-undetectable levels. 'Gestational acromegaly' probably develops to foster fetoplacental growth; its side-effects include facial oedema, carpal tunnel symptoms, and nose enlargement.

15 Puberty sees GH- and gonad-mediated rises in bone and muscle mass + other 'acromegalic' effects.

Physiology This is the passage of large volumes ($>3\text{L/day}$) of dilute urine due to impaired water resorption by the kidney, because of reduced ADH secretion from the posterior pituitary (cranial DI) or impaired response of the kidney to ADH (nephrogenic DI).

Symptoms Polyuria; polydipsia; dehydration; symptoms of hypernatraemia (p672). *Polydipsia* can be uncontrollable and all-consuming, with patients drinking anything and everything to hand: in such cases, if beer is on tap, disaster will ensue!

Causes of cranial DI • *Idiopathic* ($\leq 50\%$). • *Congenital*: defects in ADH gene, **DIDMOAD**.¹⁶ • *Tumour* (may present with DI + hypopituitarism): craniopharyngioma, metastases, pituitary tumour. • *Trauma*: temporary if distal to pituitary stalk as proximal nerve endings grow out to find capillaries in scar tissue and begin direct secretion again. • *Hypophysectomy*. • *Autoimmune* hypophysitis (p232). • *Infiltration*: histiocytosis, sarcoidosis.¹⁷ • *Vascular*: haemorrhage.¹⁸ • *Infection*: meningoencephalitis.

Causes of nephrogenic DI • *Inherited*. • *Metabolic*: low potassium, high calcium. • *Drugs*: lithium, demeclocycline. • *Chronic renal disease*. • *Post-obstructive uropathy*.

Tests U&E, Ca^{2+} , glucose (exclude DM), serum and urine osmolalities. Serum osmolality estimate $\approx 2 \times (\text{Na}^+ + \text{K}^+) + \text{urea} + \text{glucose}$ (all in mmol/L). Normal plasma osmolality is 285–295 mOsm/kg, and urine can be concentrated to more than twice this concentration. Significant DI is excluded if urine to plasma (U:P) osmolality ratio is more than 2:1, provided plasma osmolality is no greater than 295 mOsm/kg. In DI, despite raised plasma osmolality, urine is dilute with a U:P ratio < 2 . In primary polydipsia there may be dilutional hyponatraemia—and as hyponatraemia may itself cause mania, be cautious of saying ‘It’s water intoxication from psychogenic polydipsia’.

Diagnosis *Water deprivation test*: See BOX ‘The 8-hour water deprivation test’. NB: it is often difficult to differentiate primary polydipsia from partial DI. $\Delta\Delta$: DM; diuretics or lithium use; *primary polydipsia* causes symptoms of polydipsia and polyuria with dilute urine. Its cause is poorly understood;¹⁹ it may be associated with schizophrenia or mania ($\pm \text{Li}^+$ therapy), or, rarely, hypothalamic disease (neurosarcoid; tumour; encephalitis; brain injury; HIV encephalopathy). As part of this syndrome, the kidneys may lose their ability to fully concentrate urine, due to a wash-out of the normal concentrating gradient in the renal medulla.

Treatment *Cranial DI*: MRI (head); test anterior pituitary function (p232). Give desmopressin, a synthetic analogue of ADH (eg Desmomet® tablets).

Nephrogenic: Treat the cause. If it persists, try bendroflumethiazide 5mg po/24h. NSAIDs lower urine volume and plasma Na^+ by inhibiting prostaglandin synthase: prostaglandins locally inhibit the action of ADH.

→ Emergency management

- Do urgent plasma U&E, and serum and urine osmolalities. Monitor urine output carefully and check U&E twice a day initially.
- IVI to keep up with urine output. If severe hypernatraemia, do not lower Na^+ rapidly as this may cause cerebral oedema and brain injury. If Na^+ is ≥ 170 , use 0.9% saline initially—this contains 150 mmol/L of sodium. Aim to reduce Na^+ at a rate of less than 12 mmol/L per day. Use of 0.45% saline can be dangerous.
- Desmopressin 2 mcg IM (lasts 12–24h) may be used as a therapeutic trial.

16 **DIDMOAD** is a rare autosomal recessive disorder: **D**iabetes **I**nsipidus, **D**iabetes **M**ellitus, **O**ptic **A**trophy and **D**eafness (also known as Wolfram’s syndrome).

17 Suspect neurosarcoidosis if tcsf protein (seen in 34%), facial nerve palsy (25%), csf pleocytosis (23%), diabetes insipidus (21%), hemiparesis (17%), psychosis (17%), papilloedema (15%), ataxia (13%), seizures (12%), optic atrophy (12%), hearing loss (12%), or nystagmus (9%).

18 Sheehan’s syndrome is pituitary infarction from shock, eg postpartum haemorrhage. It is rare.

19 Most of us could drink 20L/d and not be hyponatraemic; some get hyponatraemic drinking 5L/d; they may have **P**sychois, **I**ntermittent hyponatraemia, and **P**olydipsia (**PIP** syndrome), ?from tnttravascular volume leading to tatrial natriuretic peptide, p137, hence natriuresis and hyponatraemia.



The 8-hour water deprivation test

Tests the ability of kidneys to concentrate urine for diagnosis of DI (dilute urine in spite of dehydration), and then to localize the cause (table 5.5). Do not do the test before establishing that urine volume is $>3\text{L/d}$ (output less than this with normal plasma Na^+ and osmolality excludes significant disturbance of water balance).

- Stop test if urine osmolality $>600\text{mOsmol/kg}$ in Stage 1 (DI is excluded).
- Free fluids until 07.30. Light breakfast at 06.30, no tea, no coffee, no smoking.

Stage 1 Fluid deprivation (0–8h): for diagnosis of DI. Start at 08.00.

- Empty bladder, then no drinks and only dry food.
- Weigh hourly. If $>3\%$ weight lost during test, order urgent serum osmolality. If $>300\text{mOsmol/kg}$, proceed to Stage 2. If <300 , continue test.
- Collect urine every 2h; measure its volume and osmolality.
- Venous sample for osmolality every 4h.
- Stop test after 8h (16.00) if urine osmolality $>600\text{mOsmol/kg}$ (ie normal).

Stage 2 Differentiate cranial from nephrogenic DI.

- Proceed if urine still dilute—ie urine osmolality $<600\text{mOsmol/kg}$.
- Give desmopressin 2mcg IM . Water can be drunk now.
- Measure urine osmolality hourly for the next 4h.

Table 5.5 Interpreting the water deprivation test

Diagnosis	Urine osmolality
Normal	Urine osmolality $>600\text{mOsmol/kg}$ in Stage 1 u:P ratio >2 (normal concentrating ability)
Primary polydipsia	Urine concentrates, but less than normal, eg $>400\text{--}600\text{mOsmol/kg}$
Cranial DI	Urine osmolality increases to $>600\text{mOsmol/kg}$ after desmopressin (if equivocal an extended water deprivation test may be tried (no drinking from 18:00 the night before))
Nephrogenic DI	No increase in urine osmolality after desmopressin

Syndrome of inappropriate ADH secretion (SIADH)

In SIADH, ADH continues to be secreted in spite of low plasma osmolality or large plasma volume. Diagnosis requires concentrated urine ($\text{Na}^+ >20\text{mmol/L}$ and osmolality $>100\text{mOsmol/kg}$) in the presence of hyponatraemia and low plasma osmolality. Causes are numerous. See p673.